Discrete Mathematics

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Chapter 1

Basics of counting

Notation. Let A be a finite set. We denote $|A|$ the cardinality of A, i.e. the number of elements in the set.

Definition 1.1. Denote by $[n]$ the set of first n natural numbers $[n] := \{1, 2, \dots, n\}$

1.1 Bijections

Theorem 1.1. *If there exists a bijection between sets A and B then $|A| = |B|$.*

1.2 Operations with finite Sets

Exercise 1.1. Suppose that you know $|A|$ and $|B|$. What can you say about $|A \cap B|$, $|A \cup B|$ and $|A \setminus B|$?

Definition 1.2 (Addition Rule)**.** If $A \cap B = \varnothing$ then $|A \cup B| = |A| + |B|$

Definition 1.3 (Cartesian Product)**.**

$$A \times B = \{(a, b) : a \in A, b \in B\}$$

Definition 1.4.

$$|A \times B| = |A| \cdot |B|$$

Definition 1.5 (Disjoint Union)**.**

$$A \sqcup B = A \times \{0\} \cup B \times \{1\}$$

Theorem 1.2.

$$|A \sqcup B| = |A| + |B|$$

Definition 1.6 (Exponential object)**.**

$$A^B := \{f : f \text{ is a function from } B \text{ to } A\}$$

Theorem 1.3.

$$|A^B| = |A|^{|B|}$$

Exercise 1.2. What is the number of n-letter words in an m-letter alphabet?

Remark 1.4. All subsets of A are bijective to the number of functions from A to $\{0, 1\}$.
Let $B \subset A$ then the function

$$B \to \mathbb{1}_{\mathbb{B}}$$
$$a \mapsto \begin{cases} 0 & \text{if } a \notin B \\ 1 & \text{if } a \in B \end{cases}$$

is called the indicative function of B.

Definition 1.7. A permutation π of a set A is a bijection $\pi : A \to A$.

Notation. The set of permutations of $[n]$ is denoted by S_n.

Theorem 1.5.

$$|S_n| = n!$$

Exercise 1.3. Construct a bijection between S_n and $[1] \times [2] \times \cdots \times [n]$.

1.3 Probabilistic method

Definition 1.8. A finite probability space is a pair (Ω, P) where Ω is a finite set and $P : 2^\Omega \to [0, 1]$ is a function assigning a number from the interval $[0, 1]$ to every subset of Ω such that:

[1]

- $P(\varnothing) = 0$
- $P(\Omega) = 1$
- $P(A \cup B) = P(A) + P(B)$ for any two disjoint sets $A, B \subseteq \Omega$

1.3.1 Probability theory to set theory dictionary

The set Ω can be thought as the set of all possible outcomes of some random experiment.

Definition 1.9. Elements of Ω are called elementary events.

Definition 1.10. Subsets of Ω are called events.

Remark 1.6. Elementary events are not events.

Definition 1.11. Let $w \in \Omega,\ A, B \subseteq \Omega$

- $w \in A \Leftrightarrow$ event A ocurred

Example 1.1 (A random sequence of 0s and 1s)**.** $\Omega = \{0, 1\}^n$ = all n-term sequences of 0s and 1s with $|\Omega| = 2^n$.

For $A \subseteq \Omega :\ P(A) := \frac{|A|}{|\Omega|} = \frac{|A|}{2^n}$.

Consider a set $A \subseteq \Omega$ defined as $A := \{(w_1, \dots, w_n) : w_1 = 1\}$. A is the event : "the first element of a random sequence is 1".

Problem 1.1. What is the probability of A?

1.4 Binomial coefficients

Definition 1.12. A binomial coefficient $\binom{n}{k}$ is the number of ways in which one can choose k objects out of n distinct objects, assuming the order of the elements does not matter.

Remark 1.7. This is the same as the number of subsets of k elements of an n-element set.

Proposition 1.8.

$$\binom{n}{k} = \frac{n!}{(n-k)!k!}$$

Notation. Let A be a finite set and k be a non-negative integer. Then $\binom{A}{k}$ is the set of k-elements of A:

$$|\binom{A}{k}| = \binom{|A|}{k}$$

Proposition 1.9. *The following identities hold:*

- $\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}$
- $\binom{n}{k}$ *is the k-th element in the n-th line of Pascal's triangle.*

Proof. Each subset of the set $[n+1]$ either contains element $n+1$ or not.

Number of $(k+1)$-element subsets of $[n+1]$ containing $n+1$ is $\binom{n}{k}$.

Number of $(k+1)$-elements subsets of $[n+1]$ not containing $n+1$ is $\binom{n}{k+1}$. ☺

Proposition 1.10. *The number of subsets of an n-element set is* 2^n*, since we have*

$$2^n = \binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n}$$

The number of subsets of an n-element set having odd cardinality is 2^{n-1}*. The number of subsets of an n-element set having even cardinality is* 2^{n-1}*.*

Proof. Consider the map $\phi : 2^{[n]} \to 2^{[n]}$ defined by

$$\phi(A) = A\Delta\{1\} = \begin{cases} A \setminus \{1\} & \text{if } 1 \in A \\ A \cup \{1\} & \text{otherwise} \end{cases}$$

Cardinalities of subsets A and $\phi(A)$ always have different parity.

Since $\phi \circ \phi = \mathrm{Id}$ we deduce that ϕ is a bijection between the set of "odd" subsets and the set of "even" subsets. ☺

Theorem 1.11 (Binomial theorem)**.**

$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \cdots + \binom{n}{n}x^n = \sum_{i=0}^{n} \binom{n}{i} x^i$$

Proposition 1.12. *Assume we have k identical objects and n different persons. Then, the number of ways in which one can distribute this k objects among the n persons equals*

$$\binom{n+k-1}{n-1} = \binom{n+k-1}{k}$$

Remark 1.13. Equivalently, it is a number of solutions to te equation $x_1 + \cdots + x_n = k$ in non-negative integers.

Proof. Let A be the set of all solutions of equation $x_1 + \cdots + x_n = k$ such that $x_i \in \mathbf{Z}_{\geq 0}$.

Let B be the set of all subsets of cardinality $n-1$ in $[n+n-1]$.

We construct the bijection $\psi : A \to B$ in the following way

$$A := (x_1, \dots, x_n) \mapsto B := \{x_1 + 1, x_1 + x_2 + 2, \dots, x_1 + \cdots + x_{n-1} + n - 1\}$$

We have $x_1 + x_2 + \cdots + x_n = k$. Consider the following string

$$\underbrace{\bullet \dots \bullet}_{x_1 \text{ bullets}} + \underbrace{\bullet \dots \bullet}_{x_2 \text{ bullets}} + \cdots + \underbrace{\bullet \dots \bullet}_{x_n \text{ bullets}}$$

This string consists of $k+n-1$ symbols : k "$\bullet$" and $n-1$ "+".

Set $B = \{$positions of "+" in the string$\}$

We show that ψ is a bijection by computing its inverse map. Let $\mathbf{b}$ be an element of B. Suppose that

$$1 \leq b_1 < b_2 < \cdots < b_{n-1} \leq k + n - 1$$

are the elements of $\mathbf{b}$ written in the increasing order. Then the preimage $\psi^{-1}(\mathbf{b})$ is an n-tuple of integers $(x_1, \dots, x_n)$ defined by

$$\begin{aligned} x_1 &= b_1 - 1 \\ x_i &= b_i - b_{i-1}, \quad i = 2, \dots, n-1 \\ x_n &= k + n - 1 - b_{n-1} \end{aligned}$$

It is easy to see from these equations that the numbers $x_i, i = 1, \dots, n$ are non-negative integeres and $x_1 + \dots + x_n = k$.

Since there is a bijection between sets A and B, their cardinalities are equal adn

$$|A| = |B| = \binom{k+n-1}{n-1}$$

☺

Definition 1.13 (Big O)**.** Let $f, g : \mathbf{Z}_{\geq 0} \to \mathbf{R}$, we say that f is big $\mathcal{O}$ of g and write $f(x) = \mathcal{O}(g(x))$ if there exists $n_0 \in \mathbf{Z}_{\geq 0}$ and $c \in \mathbf{R}$ such that for all $n > n_0$: Lecture 2

$$|f(n)| < c|g(n)|$$

Definition 1.14 (Little o)**.** Let $f, g : \mathbf{Z}_{\geq 0} \to \mathbf{R}$, we say that f is little o of g and write $f(x) = o(g(x))$ if there exists $n_0 \in \mathbf{Z}_{\geq 0}$ and $c \in \mathbf{R}$ such that for all $n > n_0$:

$$\lim_{n \to \infty} \frac{f(n)}{g(n)} = 0$$

Notation. Let $f, g : \mathbf{Z}_{\geq 0} \to \mathbf{R}$, we say that $f \sim g$ if $\lim_{n \to \infty} \frac{f(n)}{g(n)} = 1$.

Example 1.2. As n approaches ∞:

- $n = \mathcal{O}(n^2)$
- $n = o(n^2)$
- $\sin(n) = \mathcal{O}(1)$
- $n^2 \sim n^2 + 2n + 3$

1.5 Stirling's approximation

We want to approximate the function $\log(n!) = \log(1) + \dots + \log(n)$. We can attempt to do so by calculating $\int_1^n \log x \, dx = n \log n - n + 1$.

Theorem 1.14 (Stirling's approximation)**.**

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \tag{1.1}$$

Idea of the proof. For $n \in \mathbf{Z}$ we want to prove that

$$n! = \int_0^\infty x^n e^{-x} \, dx$$

to do so we can integrate the right-hand expression through the variable change $x = ny$

$$\begin{aligned} n! &= \int_0^\infty e^{n \log(x) - x} \, dx \\ &= e^{n \log(n)} n \int_0^\infty e^{n(\ln y - y)} \, dy \qquad (\star) \end{aligned}$$

to integrate $\star$ we can use Laplace's method to approximate integrals of the form

$$\int_a^b e^{nf(x)}\,dx \text{ as } n \to \infty$$

we apply a variable change $y = 1 + t$ so that

$$\begin{aligned}
\int_0^\infty e^{n(\log y - y)}\,dy &= \int_{-1}^\infty e^{n(\log(1+t)-t-1)}\,dt \\
&= e^{-n}\int_{-1}^\infty e^{n(-t^2/2+t^2\varphi(t))}\,dt && |u = \sqrt{n}t \\
&= e^{-n}\int_{-\sqrt{n}}^\infty e^{-u^2/2}e^{u^2\varphi(u/\sqrt{n})}\frac{1}{\sqrt{n}}\,du \\
&= \frac{e^n}{\sqrt{n}}\int_{-\infty}^\infty e^{-u^2/2}\,du \\
&= \frac{\sqrt{2\pi}}{\sqrt{n}e^n}
\end{aligned}$$

☺

1.6 Birthday paradox

Question. Suppose that there are 25 students in a math class. What are the chances that there is a pair of students who share the same birthday?

Answer. We imagine a list of student's birthdays

- Alice born on 2902
- Bob born on 108
- Charlotte born on 101

The number of possible lists is given by 366^{25} and the number of lists without coincidences by $366 \cdot 365 \cdot \dots \cdot 342$.

We assume that all birthdays are uniformly distributed over all 366 days and that each person's birthday can be interpreted as an independent event which means that all birthday lists are equally likely. Therefore we come up with the probability formula:

$$p = 1 - \frac{366 \cdot 365 \cdot \dots \cdot 342}{366^{25}}$$

More generally to estimate the probability of a coincidence we have the following theorem:

Theorem 1.15. *Suppose that $k \le n$ are positive integers and each of k different people chooses 1 element from the set $[n]$. Their choices are uniformly random and independent. Then the probability $P = \frac{n!}{(n-k)!n^k}$ that they have chosen k different elements can be estimated as*

$$e^{(-k(k-1))/(2(n-k+1))} \le P \le e^{(-k(k-1))/(2n)} \tag{1.2}$$

Lemma 1.16. *For $x > 0$:*

$$\frac{x-1}{x} \le \log(x) \le x - 1$$

Proof. We estimate

$$\begin{aligned}
&\log\left(\frac{n^k}{n(n-1)\dots(n-k+1)}\right)\\
&= \log\left(\frac{n}{n-1}\right) + \log\left(\frac{n}{n-2}\right) + \dots + \log\left(\frac{n}{n-k+1}\right)\\
&\leq \left(\frac{n}{n-1} - 1\right) + \left(\frac{n}{n-2} - 1\right) + \dots + \left(\frac{n}{n-k+1} - 1\right)\\
&= \frac{1}{n-1} + \frac{2}{n-2} + \dots + \frac{k+1}{n-k+1}\\
&\leq \frac{1}{n-k+1} + \frac{2}{n-k+1} + \dots + \frac{k-1}{n-k+1}\\
&= \frac{1}{n-k+1}(1 + 2 + \dots + (k-1)) = \frac{k(k-1)}{2(n_k+1)}
\end{aligned}$$

we can then apply the exponential function to both sides of our estimates to get

$$e^{(-k(k-1)/(2(n-k+1))} \leq \frac{n(n-1)\dots(n-k+1)}{n^k} \leq e^{(-k(k-1))/(2n)} \tag{1.3}$$

☺

1.7 Approximating binomial coefficients; and the bell curve

By looking at Pascal's triangle, we can observe that the middle binomial coefficients $\binom{n}{\lceil n/2 \rceil}$ and $\binom{n}{\lfloor n/2 \rfloor}$ are the biggest. Furthermore, by looking at the ratio between the growth of this coefficient and 2^n we can see that their ratio plot resembles a function of that of a logarithim function.

We can estimate this middle binomial coefficient

$$\binom{2m}{m} = \frac{(2m)!}{(m!)^2}$$

by using 1.3, we set $k = n$ to get

$$e^{\frac{-n(n-1)}{2}} n^n \leq n! \leq e^{-(n-1)/2} n^n.$$

However we can get a much better estimate by using Stirling's formula 1.1 which implies that

$$\binom{n}{n/2} \sim \sqrt{\frac{2}{\pi n}} 2^n$$

Proposition 1.17. *Let m, t be positive integers and $t \leq m$. Then*

$$e^{-t^2/(m-t+1)} \leq \frac{\binom{2m}{m-t}}{\binom{2m}{m}} \leq e^{-t^2/(m+t)}$$

Proof. We prove the lower bound. We have

$$\frac{\binom{2m}{m}}{\binom{2m}{m-t}} = \frac{(m+t)(m+t-1)\dots(m+1)}{m(m-1)\dots(m-t+1)}$$

which we can estimate

$$\begin{aligned}
&\log\left(\frac{(m+t)(m+t-1)\dots(m+1)}{m(m-1)\dots(m-t+1)}\right)\\
&=\log\left(\frac{m+t}{m}\right)+\log\left(\frac{m+t-1}{m-1}\right)+\dots+\log\left(\frac{m+1}{m-t+1}\right)\\
&\leq\left(\frac{m+t}{m}-1\right)+\left(\frac{m+t-1}{m-1}-1\right)+\dots+\left(\frac{m+1}{m-t+1}-1\right)\\
&=\frac{t}{m}+\frac{t}{m-1}+\dots+\frac{t}{m-t+1}\\
&\leq\frac{t}{m-t+1}+\frac{t}{m-t+1}+\dots+\frac{t}{m-t+1}=\frac{t^2}{m-t+1}
\end{aligned}$$

☺

1.8 Inclusion-exclusion principle

Lecture 3

Theorem 1.18 (Exclusion-inclusion). *Let $A_1,\dots,A_n$ be finite sets. Then the following holds:*

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{1\leq i\leq n}|A_i| - \sum_{1\leq i<j\leq n}|A_i\cap A_j| + \sum_{1\leq i<j<k\leq n}|A_i\cap A_j\cap A_k| - \dots + (-1)^{n-1}|A_1\cap\dots\cap A_n|$$

Lemma 1.19. *Let $B_1,\dots,B_m$ be finite sets and $w_1,\dots,w_m\in\mathbf{R}$ be some real weights. Then*

$$\sum_{i=1}^{m} w_i|B_i| = \sum_{i=1}^{m}\sum_{b\in B_i} w_i = \sum_{b\in\bigcup_i B_i}\sum_{i:b\in B_i} w_i$$

Proof of the theorem 1.18. Our goal is to show that every element of $\bigcup_i A_i$ is counted in the inclusion-exlcusion formula exactly once.

Suppose that $a\in A_1\cup A_2\cup\dots\cup A_n$ is such that it belongs to precisely k different sets $A_j, j=1,\dots,n$. This element is counted

$$\binom{k}{1}-\binom{k}{2}+\binom{k}{3}+\dots+(-1)^{n-1}\binom{k}{n}=\binom{k}{1}-\binom{k}{2}+\binom{k}{3}+\dots+(-1)^{k-1}\binom{k}{k}=1$$

time. ☺

Exercise 1.4 (Old exam question). Consider a 3×3 grid with each square colored either blue or red. Compute the total number of colorings which do not contain a 2×2 red square.

Solution. We denote S_1,S_2,S_3,S_4 the sets of colorings that contain a red square. For $j=1,\dots,4$, let A_j be the set of colorings that contain the square S_j. Therefore we can find the number of colorings without square by substracting from the total possible colorings the number of colorings with some squares which gives

$$\#\text{ of colorings without squares } = 2^9-\bigcup_{j=1}^{4} A_j$$

we then conclude by using 1.18. ☺

1.9 Applications of the formula

Question. A hat-check girl completely loses track of which of n hats belong to which owners, and hands them back at random to their n owners as the latter leave. What is the probability p_n that nobody receives their own hat back?

Reformulation: Find the number of permutations of the set $[n]$ without fixed points.

Answer. Let A be the set of all permutations and A_i be the set of permutations of the set $[n]$ for which i is a fixed point. The number of permutations with no fixed points is

$$|A| - \left|\bigcup_{i=1}^{n} A_i\right|.$$

This way $A_i \cap A_j$ is the set of permutations for which i and j are fixed points and so on:

$$\begin{aligned} |A| &= n! \\ |A_i| &= (n-1)! \\ |A_j \cap A_i| &= (n-2)! \\ &\dots \end{aligned}$$

This gives

$$\begin{aligned} |A| - \left|\bigcup_{i=1}^{n} A_i\right| &= n! - \binom{n}{1}(n-1)! + \binom{n}{2}(n-2)! - \dots \\ &= n! - \frac{n!(n-1)!}{1!(n-1)!} + \frac{n!(n-2)!}{2!(n-2)!} - \dots \\ &= n!\left(\frac{1}{0!} - \frac{1}{1!} + \frac{1}{2!} - \dots + (-1)^n \frac{1}{n!}\right) \\ &\simeq n!(\exp(-1)). \end{aligned}$$

Thus we see that the probability p_n that nobody receives their own hat back is

$$p_n = \frac{1}{0!} - \frac{1}{1!} + \frac{1}{2!} - \dots + (-1)^n \frac{1}{n!}$$

which as n approaches infinity converges to $\frac{1}{e}$.

Definition 1.15. In number theory, Euler's totient function $\phi(n)$ counts the positive integers up to a given integer n that are relatively prime to n.

Example 1.3. Among the number $\{1, 2, 3, 4, 5, 6\}$ only 1 and 5 are coprimes to 6 therefore $\phi(6) = 2$.

Example 1.4. If p is a prime number then $\phi(p) = p - 1$ and $\phi(p^k) = p^k - p^{k-1}$.

Proposition 1.20. *Suppose that a number n has the prime factorisation $n = p_1^{k_1} \dots p_m^{k_m}$ then:*

$$\phi(n) = n \prod_{i=1}^{m} \left(1 - \frac{1}{p_i}\right).$$

Proof.

- Let A be the set of all numbers in $[n]$ not coprime with n.
- Let A_i be the set of all numbers in $[n]$ divisible by p_i.

Then $A = \bigcup_{i=1}^{m} A_i$ and $|A_i| = \frac{n}{p_i}$, $|A_i \cap A_j| = \frac{n}{p_i p_j}$ and so on. By 1.18 we find:

$$\begin{aligned} \phi(n) &= n - |A| \\ &= n - \sum_{1 \le i \le n} \frac{n}{p_i} + \sum_{1 \le i < j \le m} \frac{n}{p_i p_j} - \sum_{1 \le i < j < k \le m} \frac{n}{p_i p_j p_k} + \dots = n \prod_{i=1}^{m} \left(1 - \frac{1}{p_i}\right) \end{aligned}$$

☺

1.9.1 Bonferroni's inequalities

Theorem 1.21. *Let $A_1, \dots, A_n$ be finite sets. Let $k \in \{1, \dots, n\}$ be an odd integer. Then*

$$\left|\bigcup_{i=1}^{n} A_i\right| \leq \sum_{s=1}^{k} \sum_{J \subset \{1,\dots,n\}} (-1)^{s-1} \cdot \left|\bigcap_{i \in J} A_j\right|$$

Let $\ell \in \{1, \dots, n\}$ be an even integer. Then

$$\left|\bigcup_{i=1}^{n} A_i\right| \geq \sum_{s=1}^{\ell} \sum_{J \subset \{1,\dots,n\}} (-1)^{s-1} \cdot \left|\bigcap_{j \in J} A_j\right|$$

Proof. By induction on n we verify the cases $n = 1$ and $n = 2$ because $|A_1| = |A_1|$ and $|A_2 \cup A_2| \leq |A_1| + |A_2|$. Therefore we assume the property correct for $n - 1$ sets.

Let k be an odd/even integer i $\{1, \dots, n - 1\}$:

$$\sum_{s=1}^{k} \sum_{J \subset \{1,\dots,n\}} (-1)^{s-1} \left|\bigcup_{j \in J} A_j\right| =$$

$$= \sum_{s=1}^{k} \left(\sum_{J \subset \{1,\dots,n-1\}} (-1)^{s-1} \left|\bigcap_{j \in J} A_j\right| + \sum_{J} (-1)^{s-1} \left|\bigcap_{j \in J} (A_j \cap A_n)\right| \right)$$

$$= \sum_{s=1}^{k} \sum_{J} (-1)^{s-1} \left|\bigcap_{j \in J} A_j\right| + |A_n| - \sum_{s=1}^{k-1} \sum_{J} (-1)^{s} \left|\bigcap_{j \in J} (A_j \cap A_n)\right|$$

By the induction hypothesis :

$$\sum_{s=1}^{k} \sum_{J} (-1)^{s-1} \left|\bigcap_{j \in J} A_j\right| \leq / \geq \left|\bigcup_{j=1}^{n-1} A_j\right|$$

and

$$\sum_{s=1}^{k-1} \sum_{J} (-1)^{s} \left|\bigcap_{j \in J} (A_j \cap A_n)\right| \geq / \leq \left|\bigcup_{j=1}^{n-1} (A_j \cap A_n)\right|$$

Therefore we arrive at

$$\sum_{s=1}^{k} \sum_{J} (-1)^{s-1} \left|\bigcap_{j \in J} A_j\right| = \sum_{s=1}^{k} \sum_{J} (-1)^{s-1} \left|\bigcap_{j \in J} A_j\right| + |A_n| - \sum_{s=1}^{k-1} \sum_{J} (-1)^{s} \left|\bigcap_{j \in J} (A_j \cap A_n)\right|$$

We use the induction hypothesis

$$\leq / \geq \left|\bigcup_{j=1}^{n-1} A_j\right| + |A_n| - \left|\left(\bigcup_{j=1}^{n-1}\right) \cap A_n\right|.$$

By 1.18 this number equals:

$$\left|\left(\bigcup_{j=1}^{n}\right) \cup A_n\right| = \left|\bigcup_{j=1}^{n} A_j\right|$$

☺

1.10 Generating functions: combinatorial applications of polynomials

Lecture 4

Example 1.5. How many ways are there to pay the amount of 21 CHF with :

- 6 1CHF coins
- 5 2CHF coins
- 4 5CHF coins

The number in question is the numbers of solutions of:

$$x_1 + x_2 + x_3 = 21 \tag{1.4}$$

Where

$$x_1 = \{0, 1, \dots, 6\}, \quad x_2 = \{0, 2, \dots, 10\}, \quad x_3 = \{0, 5, \dots, 20\}$$

Consider the following polynomials

$$\begin{aligned} p_1(x) &= 1 + x + x^2 + x^3 + x^4 + x^5 + x^6 \\ p_2(x) &= 1 + x^2 + x^4 + x^6 + x^8 + x^{10} \\ p_3(x) &= 1 + x^5 + x^{10} + x^{15} + x^{20} \end{aligned}$$

The number of solutions of 1.4 is the coefficient of x^{21} in the product $p_1 p_2 p_3$.

Operations with polynomials

Let $p(x) = \sum_{k=0}^{n} a_k x^k$ and $q(x) = \sum_{k=0}^{n} b_k x^k$ two polynomials.

- Addition: $(p + q)(x) = \sum_{k=1}^{n} (a_k + b_k) x^k$
- Multiplication: $p(x)q(x) = \sum_{k=1}^{2n} (a_0 b_k + a_1 b_{k-1} + \dots + a_k b_0) x^k$
- Substitution: $p(q(x)) = a_0 + a_1 q(x) + a_2 q(x)^2 + \dots + a_n q(x)^n$
- Differentiation: $\frac{d}{dx} p(x) = \sum_{k=1}^{n} a_k k x^{k-1} = \sum_{k=0}^{n-1} (k+1) a_{k+1} x^k$
- Integration: $\int_0^x p(y)\, dy = \sum_{k=0}^{n} a_k \frac{x^{k+1}}{k+1} = \sum_{k=1}^{n+1} \frac{a_{k-1}}{k} x^k$

Identities with binomial coefficients

Example 1.6. Prove the identity

$$\sum_{i=0}^{n} \binom{n}{i}^2 = \binom{2n}{n}$$

We consider the identity

$$(1 + x)^n (1 + x)^n = (1 + x)^{2n}$$

and we can compute the coefficient of x^n on the left and right hand side of this equality.

On the left:

$$(1 + x)^n (1 + x)^n = (\binom{n}{0} + \binom{n}{1} x + \dots + \binom{n}{n} x^n)(\binom{n}{0} + \binom{n}{1} x + \dots + \binom{n}{n} x^n)$$

which gives the coefficient of x^n

$$\binom{n}{0}\binom{n}{n} + \binom{n}{1}\binom{n}{n-1} + \dots + \binom{n}{n}\binom{n}{0} = \binom{n}{0}^2 + \binom{n}{1}^2 + \dots + \binom{n}{n}^2$$

On the right hand side

$$(1 + x)^{2n} = \binom{2n}{0} + \binom{2n}{1} x + \dots + \binom{2n}{2n} x^{2n}$$

Example 1.7. Compute

$$\sum_{k=0}^{n} k \binom{n}{k}$$

Consider the polynomial

$$p(x) = (1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k$$

We diffrentiate $p(x)$:

$$\frac{d}{dx} p(x) = \frac{d}{dx}[(1+x)^n] = n(1+x)^{n-1}$$

On the other hand

$$\frac{d}{dx} p(x) = \frac{d}{dx}\left[\sum_{k=0}^{n} \binom{n}{k} x^k\right] = \sum_{k=1}^{n} k \binom{n}{k} x^{k-1}$$

Thus

$$\sum_{k=1}^{n} k \binom{n}{k} x^{k-1} = n(1+x)^{n-1}$$

and we replace $x = 1$ to obtain the desired computation

$$\sum_{k=1}^{n} k \binom{n}{k} = 2^{n-1} n$$

Theorem 1.22 (Multinomial theorem)**.** *The following holds*

$$(x_1 + x_2 + \dots + x_n)^k = \sum_{\substack{i_1, i_2, \dots, i_n \geq 0 \\ i_1 + i_2 + \dots + 1_n = k}} \frac{k!}{i_1! i_2! \dots i_n!} x_1^{i_1} x_2^{i_2} \dots x_n^{i_n}$$

Proof.

$$\frac{k!}{i_1! \dots i_{n!}}$$

is the number of sequences of length k from letters $x_1, x_2, \dots, x_n$ such that each letter x_j is used i_j-times.

We prove by induction: We pose the basis of induction as the case $n = 2$, which is the binomial theorem, and assume the property holds for n. So for $n + 1$:

$$\begin{aligned}
((x_1 + \dots + x_n) + x_{n+1})^k &= \sum_{i=0}^{k} \binom{k}{i} (x_1 + \dots + x_n)^{k-i} x_{n+1}^i \\
&= \sum_{i=0}^{k} \left(\sum_{\substack{i_1, \dots, i_n \geq 0 \\ i_1 + i_2 + \dots + 1_n = k-i}} \frac{(k-1)!}{i_1! \dots i_{n!}} x_1^{i_1} \dots x_n^{i_n} \right) \binom{k}{i} x_{n+1}^i \\
&= \sum_{\substack{i_1, \dots, i_{n+1} \geq 0 \\ i_1 + \dots i_{n+1} = k}} \frac{k!}{i_1! \dots i_n! i_{n+1}!} x_1^{i_1} \dots x_n^{i_n} x_{n+1}^{i_{n+1}}
\end{aligned}$$

☺

Example 1.8. *Another proof of 1.18.* Let $A_1, \dots, A_n$ be finite sets and $U_i = \bigcup_{i=1}^{n} A_i$. Let $A \subset U$ be a subset, the indicator function of A is

$$\begin{aligned}
\mathbb{1}_A &: U \to \{0, 1\} \\
u &\mapsto \begin{cases} 1 & \text{if } u \in A \\ 0 & \text{if } u \notin A, u \in U \end{cases}
\end{aligned}$$

we use the properties:

1. $|A| = \sum_{u \in U} \mathbb{1}_A(u)$
2. $\mathbb{1}_{U \setminus A}(u) = 1 - \mathbb{1}_A(u)$
3. $\mathbb{1}_{A \cap B}(u) = \mathbb{1}_A(u) - \mathbb{1}_B(u)$

Let

$$\bigcup_{i=1}^{n} A_i = U \setminus \left(\bigcap_{i=1}^{n} (U \setminus A_i) \right)$$

We calculate the cardinality

$$\begin{aligned}
\left| \bigcup_{i=1}^{n} A_i \right| &\overset{1}{=} \sum_{u \in U} \mathbb{1}_{\bigcup_{i=1}^{n} A_i}(u) \\
&\overset{2}{=} \sum_{u \in U} \left(1 - \mathbb{1}_{\bigcap_{i=1}^{n} (U \setminus A_i)}(u) \right) \\
&\overset{3}{=} \sum_{u \in U} \left(1 - \prod_{i=1}^{n} (\mathbb{1}_{U \setminus A_i}(u)) \right) \\
&\overset{2}{=} \sum_{u \in U} \left(1 - \prod_{i=1}^{n} (1 - \mathbb{1}_{A_i}(u)) \right) \\
&= \sum_{u \in U} \left(1 - \sum_{k=0}^{n} (-1)^k \sum_{\{i_1, \dots, i_k\}} \mathbb{1}_{A_{i_1} \cap \dots \cap A_{i_k}}(u) \right) \\
&\overset{1}{=} \sum_{u \in U} \sum_{k=0}^{n} (-1)^{k+1} \sum_{I \subset \{1, \dots, n\} : |I| = k} \mathbb{1}_{\bigcap_{i \in I} A_i}(u) \\
&= \sum_{k=1}^{n} (-1)^{k+1} \sum_{I} \left| \bigcap_{i \in I} A_i \right|
\end{aligned}$$

for $u \in U$ we have

$$\begin{aligned}
\prod_{i=1}^{n} (1 - \mathbb{1}_{A_i}(u)) &= \sum_{k=0}^{n} (-1)^k \sum_{\{i_1, \dots, i_k\} \subset \{1, \dots, n\}} \prod_{s=1}^{k} \mathbb{1}_{A_{i_s}}(u) \\
&\overset{3}{=} \sum_{k=0}^{n} (-1)^k \sum_{\{i_1, \dots, i_k\}} \mathbb{1}_{A_{i_1} \cap \dots \cap A_{i_k}}(u)
\end{aligned}$$

☺

1.11 Generating functions: calculations with power series

Suppose that $a_0, \dots, a_n$ is an interesting sequence of numbers. Then $a(x) = a_n x^n + \dots + a_0$ is an interesting power series. We can see this $a(x)$ as

- A formal power series if x is considered as a formal variable.
- A power series converging for $|x| < r_0$ then $a(x)$ can be viewed as a functionn $(-r_0, r_0)$ or on a disk $\{z \in \mathbf{C} : |z| < r_0\}$

Definition 1.16. Let $\{a_n\}_{n=0}^{\infty}$ be a sequence of complex numbers. Then the generating series of this sequence is the power series

$$a(x) := \sum_{n=0}^{\infty} a_n x^n$$

Example 1.9 (Geometric series)**.** Consider the sequence $a_n = 1$ for $n = 0, \dots, \infty$. The generating series of this sequence is $a(x) = 1 + x + \dots + x^n$ which converges to $\frac{1}{1-x}$ if $|x| < 1$.

Definition 1.17 (Formal power series)**.** A formal power series is an infinite sum

$$a(x) = \sum_{n=0}^{\infty} a_n x^n$$

where $\{a_n\}_{n=0}^{\infty}$ is a sequence of complex numbers and x is a formal variable.

Proposition 1.23. *Let $a(x) = \sum_{n=0}^{\infty} a_n x^n$ be a formal power series. Suppose that there exists a positive real number K such that $|a_n| < K^n$ for all $n \in \mathbf{Z}_{\geq 0}$. Then the series $a(x)$ converges absolutely when x is substituted by a real number in the interval $\left(-\frac{1}{k}, \frac{1}{k}\right)$.*

Moreover, the function $a : \left(-\frac{1}{k}, \frac{1}{k}\right) \to \mathbf{C}$ has derivatives of all orders at 0 and $a_n = \frac{a^{(n)}(0)}{n!}$.

Operations with sequences and their generating functions

1. Addition:
$$\sum_{n=0}^{\infty} a_n x^n + \sum_{n=0}^{\infty} b_n x^n = \sum_{n=0}^{\infty} (a_n + b_n) x^n$$

2. Multiplication:
$$\left(\sum_{n=0}^{\infty} a_n x^n\right)\left(\sum_{n=0}^{\infty} b_n x^n\right) = \sum_{n=0}^{\infty}\left(\sum_{k=0}^{n} a_k b_{n-k}\right) x^n$$

3. Substitution:
$$a(b(x)) = \sum_{n=0}^{\infty} a_n b(x)^n = \sum_{n=0}^{\infty} a_n \left(\sum_{m=0}^{\infty} b_m x^m\right)^n$$

4. Differentiation:
$$\frac{d}{dx}\left(\sum_{n=0}^{\infty} a_n x^n\right) = \sum_{n=1}^{\infty} n a_n x^{n-1} = \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n$$

5. Integration:
$$\int_0^x \left(\sum_{n=0}^{\infty} a_n y^n\right) dy = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}$$

Example 1.10. What is the generating function for $a_n = n + 1$.

$$1 + 2x + \dots + nx^{n-1} = \frac{d}{dx}(x + x^2 + \dots + x^n) = \frac{d}{dx}\left(\frac{x}{1-x}\right) = \frac{1}{(1-x)^2}$$

Example 1.11. Compute the sum $\sum_{k=0}^{m} (-1)^k \binom{n}{k}$.

$$\begin{aligned}\sum_{m=0}^{\infty}\left(\sum_{k=0}^{m} (-1)^k \binom{n}{k}\right) x^k &= \sum_{k=0}^{\infty} (-1)^k \binom{n}{k}\left(\sum_{m=k}^{\infty} x^m\right) \\ &= \sum_{k=0}^{\infty} (-1)^k \binom{n}{k} \frac{x^k}{1-x} \\ &= \frac{(x-1)^n}{1-x} = -(x-1)^{n-1}\end{aligned}$$

Thus $\sum_{k=0}^{m} (-1)^k \binom{n}{k} = -\binom{n-1}{m}$.

Theorem 1.24 (Generalized binomial theorem)**.** *For every $r \in \mathbf{R}$ and every integer $k \geq 0$, let*

$$\binom{r}{k} = \frac{r(r-1)\dots(r-k+1)}{k!}$$

then the following holds

$$(1+x)^r = \binom{r}{0} + \binom{r}{1} x + ...$$

for all x with $|x| < 1$.

1.12 Binary trees and Catalan numbers

Lecture 5

Theorem 1.25. *Let $r \in \mathbf{R}$. Then the function $(1+x)^r$ has a Taylor expansion*

$$(1+x)^r = \sum_{k=0}^{\infty} \binom{r}{k} x^k$$

for $x \in (-1, 1)$.

Question. Compute $\binom{1/3}{2}$.

Answer. $\binom{1/3}{2} = -\frac{1}{9}$.

Definition 1.18 (Binary tree)**.** An inductive definition of a binary tree is given as follows:

A binary tree is either empty, or consists of one distinguished vertex called the root, plus an ordered pair of binary trees called the left subtree and the right subtree.

Let b_n denote the number of binary trees with n vertices. Our goal is to find a formula for b_n.

Example 1.12.

$$b_0 = 1 \quad b_1 = 1 \quad b_2 = 2$$

The inductive definition of a binary tree implies the following recursive formula for b_n:

$$b_n = b_0 b_{n-1} + b_1 b_{n-2} + \cdots + b_{n-1} b_0$$

Consider the generating function

$$b(x) = \sum_{k=0}^{\infty} b_n x^n$$

The recursive formula

$$b_n = \sum_{k=0}^{n-1} b_k b_{n-k-1}$$

implies

$$\begin{aligned}
b(x)b(x) &= \left(\sum_{n=0}^{\infty} b_n x^n\right)\left(\sum_{n=0}^{\infty} b_n x^n\right) \\
&= \sum_{k=0}^{\infty} \left(\sum_{m=0}^{k} b_m b_{k-m}\right) x^k \\
&= \sum_{k=0}^{\infty} b_{k+1} x^k \\
&= \frac{1}{x}\left(\sum_{k=1}^{\infty} b_k x^k\right) = \frac{1}{x}(b(x) - b_0)
\end{aligned}$$

Thus $b(x)$ satisfies the identity

$$x b^2(x) - b(x) + 1 = 0$$

which has two solutions

$$b_{1,2}(x) = \frac{1 \pm \sqrt{1-4x}}{2x}$$

of which we will consider

$$\tilde{b}(x) = \frac{1-\sqrt{1-4x}}{2x} = \frac{2}{1+\sqrt{1-4x}}$$

that has a Taylor expansion around $x = 0$

$$\tilde{b}(x) = \sum_{n=0}^{\infty} \tilde{b}_n x^n$$

and as $\tilde{b}(x)$ satisfies the equation

$$x\tilde{b}(x)^2 - \tilde{b}(x) + 1 = 0 \Rightarrow \tilde{b}(0) = b_0 = 1$$

which gives the sequence $\tilde{b}_n$:

$$\tilde{b}_n = \tilde{b}_0\tilde{b}_{n-1} + \tilde{b}_1\tilde{b}_{n-2} + \dots + \tilde{b}_{n-1}\tilde{b}_0$$

We have

$$\begin{gathered} b_0 = 1 = 1 = \tilde{b}_0 \\ b_1 = b_0^2 = \tilde{b}_0^2 = \tilde{b}_2 \\ b_2 = b_0 b_1 + b_1 b_0 = b_0 b_1 + b_1 b_0 = \tilde{b}_2 \\ \vdots \end{gathered}$$

Therefore $b_n = \tilde{b}_n$.

To compute $\tilde{b}_n$ we use the generalized binomial theorem, which implies

$$\sqrt{1-4x} = \sum_{n=0}^{\infty} (-4)^n \binom{1/2}{n} x^n$$

Thus

$$\tilde{b}(x) = \sum_{n=1}^{\infty} -\frac{1}{2}(-4)^n \binom{1/2}{n} x^{n-1}$$

therefore

$$b_n = -\frac{1}{2}(-4)^{n+1} \binom{1/2}{n+1}$$

.

Definition 1.19. The numbers b_n are called Catalan numbers.

Exercise 1.5. Show that

$$b_n = \frac{1}{n+1}\binom{2n}{n}$$

1.12.1 More Catalan numbers

Definition 1.20 (Cutting polygons into triangles)**.** A diagonal triangulation of a polygon is a partition of the polygon into triangles by non-intersecting diagonals.

Lemma 1.26. *Let $n \geq 3$. The number of diagonal triangulations a convex n-gon is the Catalan number b_{n-2}.*

Definition 1.21 (Parentheses)**.** A regular bracket structure satisfies the following two conditions:

1. The number of left and right brackets is the same.
2. The number of left brackets in any starting segment is not less than then number of right brackets.

Lemma 1.27. *The number of regular bracket structures with n pairs of brackets is the Catalan number b_n.*

Definition 1.22 (Ups and downs)**.** A Dyck path is a continuous broken line consisting of vectors $(1,1)$ and $(1,-1)$ starting at the origin $A = (0,0)$ and ending at the x-axis and always staying in the upper half-plane.

Lemma 1.28. *The number of Dyck paths consisting of $2n$ steps is the Catalan number b_n.*

1.13 Fibonnaci numbers and the Golden Ratio

Definition 1.23. The Fibonnaci sequence $\{F_n\}_{n=0}^{\infty}$ is defined by the following recursive formula:

$$F_0 = 0, \quad F_1 = 1, \quad F_{n+1} = F_n + F_{n-1}, \quad n \geq 1$$

Another interpretation of Fibonnaci numbers: Let S_n denote the number of ways one can climb n stairs if allowed to jump 1 or 2 stairs at once. S_n is the number of solutions to the equation $x_1 + x_2 + \cdots + x_k = n$ where $x_i \in \{1,2\}$ and k is not fixed. Thus we observe that

$$S_1 = 1, \quad S_2 = 2, \quad S_n = S_{n-1} + S_{n-2}$$

and $S_n = F_{n+1}$.

Definition 1.24 (Golden ratio)**.** The golden ration is the number

$$\varphi = \frac{1+\sqrt{5}}{2} \approx 1.61083\ldots$$

Proposition 1.29.

$$\lim_{n\to\infty} \frac{F_{n+1}}{F_n} = \varphi$$

1.13.1 Explicit formula for Fibonnaci numbers

Consider the generating function of the Fibonnaci sequence :

$$F(x) = F_0 + F_1 x + \cdots + F_n x^n + \ldots$$

then

$$x^2 F(x) = \sum_{n=2}^{\infty} F_{n-2} x^n, \quad xF(x) = \sum_{n=1}^{\infty} F_{n-1} x^n$$

thus

$$\begin{aligned} F(x) - xF(x) - x^2 F(x) &= \sum_{n=0}^{\infty} F_n x^n - \sum_{n=1}^{\infty} F_{n-1} x^n - \sum_{n=2}^{\infty} F_{n-2} x^n \\ &= F_0 + xF_1 - xF_0 + \sum_{n=2}^{\infty} (F_n - F_{n-1} - F_{n-2}) x^n \\ &= x \end{aligned}$$

therefore

$$F(x)(1 - x - x^2) = x$$

thus

$$F(x) = \frac{x}{1-x-x^2}$$

as a formal power series.

The function $\frac{x}{1-x-x^2}$ has derivatives of all orders at the point $x = 0$. We will compute the Taylor expansion of this function at $x = 0$.

The polynomial $1 - x - x^2$ has two roots

$$x_{1,2} = \frac{-1 \pm \sqrt{5}}{2}$$

therefore

$$\begin{aligned}
\frac{x}{1-x-^2} &= -\frac{x}{(x-x_1)(x-x_2)} \\
&= \left(\frac{1}{x-x_1} - \frac{1}{x-x_2}\right)\frac{x}{x_2-x_1} \\
&= \frac{x}{\sqrt{5}}\left(-\frac{1}{x_1}\cdot\frac{1}{1-\frac{x}{x_1}} + \frac{1}{x_2}\cdot\frac{1}{1-\frac{x}{x_2}}\right) \\
&= \frac{x}{\sqrt{5}}\left(\sum_{n=0}^{\infty} -\frac{1}{x_1}\left(\frac{x}{x_1}\right)^n + \sum_{n=0}^{\infty}\frac{1}{x_2}\left(\frac{x}{x_2}\right)^n\right) \\
&= \frac{1}{\sqrt{5}}\sum_{n=0}^{\infty}(-x_1^{-n-1} + x_2^{-n_-1})x^{n+1} \\
&= \sum_{n=1}^{\infty}\frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right)x^n
\end{aligned}$$

Therefore the Fibonnaci numbers satisfy

$$F_n = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right) = \frac{1}{\sqrt{5}}(\varphi^n - (-\varphi)^{-n})$$

for $n \geq 0$. Lecture 6

1.14 Linear recurrence relations

Definition 1.25. A sequence of complex numbers $\{a_n\}_{n=0}^{\infty}$ satisfies a linear recurrence relation if there exist numbers $c_0, c_1, \dots, c_{k-1}$ such that

$$a_{n+k} = c_0 a_n + c_1 a_{n+1} + \dots + c_{k-1}a_{n+k-1}$$

Example 1.13. The Fibonacci numbers satisfy a linear recurrence relation.

Theorem 1.30. *Suppose that a sequence $\{a_n\}_{n=0}^{\infty}$ satisfies a linear recurrence relation. Let $\lambda_1, \dots, \lambda_s$ be the complex roots of the polynomial*

$$x^k - c_{k-1}x^{k-1} - \dots - c_1 x - c_0 = 0$$

Then there exist polynomials $P_1, \dots, P_s$, where each P_i is a polynomial of degree $k_i - 1$ and

$$a_n = \sum_{i=1}^{s} P_i(n)\lambda_i^n, \quad n \in \mathbf{Z}_{\geq 0}$$

Proof. Suppose that a sequence $\{a_n\}_{n=0}^{\infty}$ satisfies a linear recurrence relation:

$$a_{n+k} = c_0 a_n + c_1 a_{n+1} + \cdots + c_{k-1} a_{n+k-1} \tag{1.5}$$

Let $a(x) = \sum_{n=0}^{\infty} a_n x^n$ be the generating function for the sequence $\{a_n\}_{n=0}^{\infty}$. The recurrence relation 1.5 implies

$$\begin{aligned}
0 &= \sum_{n=0}^{\infty} (a_{n+k} - c_{k-1} a_{n+k-1} - \cdots - c_1 a_{n+1} - c_0 a_n) x^n \\
&= \sum_{n=k}^{\infty} a_n x^{n-k} - c_{k-1} \sum_{n=k-1}^{\infty} a_n x^{n-k+1} - \cdots - c_1 \sum_{n=1}^{\infty} a_n x^{n-1} - c_0 \sum_{n=0}^{\infty} a_x x^n \\
&= x^{-k} \left(a(x) - \sum_{n=0}^{k-1} a_n x^n \right) - c_{k-1} x^{-k+1} \left(a(x) - \sum_{n=0}^{k-2} a_n x^n \right) - \cdots - c_1 x^{-1} (a(x) - a_0) - c_0 a(x)
\end{aligned}$$

Which implies

$$a(x)(x^{-k} - c_{k-1} x^{-k+1} - \cdots - c_1 x^{-1} - c_0) = \sum_{n=1}^{k} b_n x^{-}$$

so

$$a(x) = \frac{b_1 x^{-1} + b_2 x^{-2} + \cdots + b_k x^{-k}}{x^{-k} - c_{k-1} x^{-k+1} - \cdots - c_1 x^{-1} - c_0}$$

Lemma 1.31. *Let $f(x) = \frac{p(x)}{q(x)}$ be a rational function, where $p(x)$ and $q(x)$ are polynomials and* $\deg q > \deg p$.

We suppose that $q(x) = (x - \mu_1)^{\ell_1} \cdots (x - \mu_t)^{\ell_t}$ for some $\mu_1, \dots, \mu_t \in \mathbf{C}$ and $\ell_1, \dots, \ell_t \in \mathbf{Z}_{\geq 0}$. Then there exist numbers $A_{j,m}$ where $j = 1, \dots, t$ and $m = 1, \dots, \ell_j$ such that

$$f(x) = \sum_{j=1}^{t} \sum_{m=1}^{\ell_j} \frac{A_{j,m}}{(x - \mu_j)^m}$$

We assume that 1.31 is true, therefore $a(x) = \frac{p(x)}{q(x)}$ by the lemma. Suppose that $q(x) = (x - \mu_1)^{\ell_1} \cdots (x - \mu_t)^{\ell_t}$, by the lemma

$$a(x) = \sum_{j=1}^{t} \sum_{m=1}^{\ell_j} \frac{A_{j,m}}{(x - \mu_j)^m}$$

Each tern in this sum has the Taylor expansion around 0:

$$\frac{1}{(x-\mu)^m} = \frac{(-1)^{m-1}}{(m-1)!} \frac{d^{m-1}}{dx^{m-1}} \left[\frac{1}{(x-\mu)} \right] = \frac{(-1)^{m-1}}{(m-1)!} \frac{d^{m-1}}{dx^{m-1}} \left[\mu^{-1} \sum_{n=0}^{\infty} - \left(\frac{x}{\mu} \right)^n \right]$$

which gives

$$\frac{1}{(x-\mu)^m} = \alpha_m \sum_{n=0}^{\infty} -n(n-1) \cdots (n-m+2) \mu^{-n-1} x^{n-m+1}$$

Since λ_j are the roots of

$$x^k - c_{k-1} x^{k-1} - \cdots - c_1 x - c_0 = 0$$

and μ_j are the roots of

$$x^{-k} - c_{k-1} x^{-k+1} - \cdots - c_1 x^{-1} - c_0 = 0$$

therefore $\lambda_j = \mu_j^{-1}$.

If m is fixed and n is considered as a variable then

$$-n(n-1) \cdots (n-m+2)$$

is a polynomial of degree $m-1$.

We have

$$a(x) = \sum_{j=1}^{t} \sum_{m=1}^{\ell_t} \frac{A_{j,m}}{(x-\mu_j)^m}$$

where

$$\begin{aligned}\sum_{m=1}^{\ell} \frac{A_m}{(x-\mu)^m} &= \sum_{n=0}^{\infty} \left(\sum_{m=1}^{\ell} A_m \alpha_m (n+1)(n+2)\cdots(n+m-1)\lambda^m \right) \lambda^n x^n \\ &= \sum_{n=0}^{\infty} P(n)\lambda^n x^n\end{aligned}$$

with $\deg P = \ell - 1$. ☺

Example 1.14. The Fibonacci numbers $\{F_n\}_{n=0}^{\infty}$ satisfy $F_{n+2} = F_{n+1} + F_n$ for $n \in \mathbf{Z}_{\geq 0}$. The polynomial

$$x^2 - x - 1 - 0$$

has roots

$$\lambda_1 = \frac{1+\sqrt{5}}{2}, \quad \lambda_2 = \frac{1-\sqrt{5}}{2}$$

Therefore, the theorem implies that there exist complex numbers p_1 and p_2 such that

$$F_n = p_1 \left(\frac{1+\sqrt{5}}{2} \right)^n + p_2 \left(\frac{1-\sqrt{5}}{2} \right), \quad n \in \mathbf{Z}_{\geq 0}$$

The conditions $F_0 = 0$ and $F_1 = 1$ imply that

$$p_1 = \frac{1}{\sqrt{5}}, \quad p_2 = -\frac{1}{\sqrt{5}}$$

Proof of lemma 1.31. We prove the lemma by induction on t. For the base of our induction : $t = 1$ and $q(x) = (x-\mu)^\ell$ so the polynomial p can be written as

$$p(x) = \sum_{m=0}^{\ell-1} \underbrace{\frac{P^{(m)}(\mu)}{m!}}_{:=A_m} (x-\mu)^m$$

so

$$\begin{aligned}\frac{p(x)}{q(x)} &= \frac{A_0 + A_1(x-\mu) + \cdots + A_{\ell-1}(x-\mu)^{\ell-1}}{(x-\mu)^\ell} \\ &= \frac{A_0}{(x-\mu)^\ell} + \frac{A_1}{(x-\mu)^{\ell-1}} + \cdots + \frac{A_{\ell-1}}{(x-\mu)} \quad \checkmark\end{aligned}$$

We suppose the theorem holds for polynomials q with at most $t-1$ distinct roots. Consider a polynomial $q(x) = (x-\mu_1)^{\ell_1} \cdots (x-\mu_t)^{\ell_t}$. Let $q_1(x) = (x-\mu_1)^{\ell 1} \cdots (x-\mu_{t-1})^{\ell_{t-1}}$. We claim that there exist polynomials p_1, p_2 such that $\deg p_1 < \deg q_1$ and $\deg p_2 < \ell_t$ and

$$\frac{p(x)}{q(x)} = \frac{p(x)}{q_1(x)(x-\mu_t)^{\ell_t}} = \frac{p_1(x)}{q_1(x)} + \frac{p_2(x)}{(x-\mu_t)^{\ell_t}}.$$

To prove this claim, we assume without loss of generality that $\mu_t = 0$ and we define $\ell := \ell_t$. Then our claim is equivalent to the following statement :

Claim 1.32. *Let $p(x), q_1(x)$ be polynomials such that $\deg p < \deg q_1 + \ell$ and $q_1(0) \neq 0$. Then there exist polynomials p_1, p_2 such that $\deg p_1 < \deg q_1$ and $\deg p_2 < \ell$ and*

$$p(x) = x^\ell p_1(x) + q_1(x)p_2(x).$$

Proof. Our claim will follow from the division with reminder for polynomials in one variable.

Let $n := \deg q_1(x)$. We set

$$\tilde{p}(x) = x^{n+\ell-1} p\left(\frac{1}{x}\right), \quad \tilde{q_1}(x) = x^n q_1\left(\frac{1}{x}\right)$$

where $\tilde{p}$ and $\tilde{q_1}$ are polynomials and $\deg \tilde{p} \leq n+\ell-1$. Since $q_1(0) \neq 0$, we see that $\deg \tilde{q_1} = \deg q_1 = n$.

The division with remainder for univariate polynomials implies that there exist polynomials $\tilde{s}$ and $\tilde{r}$ such that $\deg \tilde{s} = \deg \tilde{p} - \deg \tilde{q_1}$ and $\deg \tilde{r} < \deg \tilde{q_1}$ and

$$\tilde{p}(x) = \tilde{q_1}(x)\tilde{s}(x) + \tilde{r}(x)$$

We set $y = \frac{1}{x}$ and obtain

$$y^{n+\ell-1}\tilde{p}\left(\frac{1}{y}\right) = y^{n+\ell-1}\tilde{q_1}\left(\frac{1}{y}\right)\tilde{s}\left(\frac{1}{y}\right) + y^{n+\ell-1}\tilde{r}\left(\frac{1}{y}\right)$$

therefore

$$p(y) = \underbrace{y^{\ell-1}\tilde{s}\left(\frac{1}{y}\right)}_{p_2} q_1(y) + y^{\ell}\,\underbrace{y^{n-1}\tilde{r}\left(\frac{1}{y}\right)}_{p_1}$$

where $\deg p_2 \leq \ell - 1$ and $\deg p_1 \leq n - 1$. ☺

The proof of the claim completes the inductions step in the lemma

$$\frac{p(x)}{q(x)} = \frac{p(x)}{q_1(x)(x-\lambda_t)^{\ell_t}} = \frac{p_1(x)}{q_1(x)} + \frac{p_2(x)}{(x-\lambda_t)^{\ell_t}}$$

which by induction hypothesis equals

$$\sum_{j=1}^{t-1}\sum_{m=1}^{\ell_j}\frac{A_{j,m}}{(x-\lambda_j)^m} + \sum_{m=1}^{\ell_t}\frac{A_{t,m}}{(x-\lambda_t)^m}$$

☺

1.14.1 Linear recurrences in the matrix form

Let $\{a_n\}_{n=0}^{\infty}$ be a sequence satisfying a linear recurrence relation 1.5. For each $n \geq 0$ we consider the vector

$$\boldsymbol{a}_n = \begin{pmatrix} a_n \\ a_{n+1} \\ \vdots \\ a_{n+k-1} \end{pmatrix}$$

then the condition 1.5 is equivalent to

$$\begin{pmatrix} a_{n+1} \\ a_{n+2} \\ \vdots \\ a_{n+k-1} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & & & \vdots \\ c_0 & c_1 & c_2 & \cdots & c_{k-1} \end{pmatrix} \begin{pmatrix} a_{n+1} \\ a_{n+2} \\ \vdots \\ a_{n+k-1} \end{pmatrix}$$

Example 1.15. Let $\{a_n\}_{n=0}^{\infty}$ be a sequence. Suppose that $a_0 = 0$, $a_1 = 1$ and $a_n = 3a_{n-1} + 2a_{n-2}$. Find an explicit formula for a_n for $n \in \mathbf{Z}$.

We make the assumption that $a_n = \lambda^n$ for $\lambda \neq 0$. Therefore

$$\begin{aligned} \lambda^n &= 3\lambda^{n-1} + 2\lambda^{n-2} \\ \Leftrightarrow 0 &= \lambda^2 - 3\lambda + 2 \end{aligned}$$

which has two roots $\lambda_1 = 1$ and $\lambda_2 = 2$. There exist $p_1, p_2 \in \mathbf{C}$ such that $a_n = p_1\lambda_1^n + p_2\lambda_2^n$. We have that

$$\begin{cases} p_1 + p_2 & = 0 \\ p_1 + 2p_2 & = 1 \end{cases}$$

then $p_1 = -1$ and $p_2 = 1$ so $a_n = -1 + 2^n$.

Example 1.16. Let $\{a_n\}_{n=0}^{\infty}$ be a sequence. Suppose that $a_0 = 0$, $a_1 = 1$ and $a_n = 2a_{n-1} - a_{n-2}$. Find an explicit formula for a_n for $n \in \mathbf{Z}$.

We make the assumption that $a_n = \lambda^n$ for $\lambda \neq 0$. So $\lambda^2 - 2\lambda + 1 = 0$ and $\lambda_{1,2} = 1$. Then $a_n = p(n)1^n$ where $\deg p = 1$. There exist $p_0, p_1 \in \mathbf{C}$ such that $a_n = p_0 + np_1$ so

$$\begin{cases} p_0 + 0p_1 & = 0 \\ p_0 + 1p_1 & = 1 \end{cases}$$

therefore $p_0 = 0$ and $p_1 = 1$ so $a_n = n$.

1.15 Mobius inversion formula

Let $f : \mathbf{Z}_{\geq 1} \to \mathbf{C}$ be a function. We define a new function $F : \mathbf{Z}_{\geq 1} \to \mathbf{C}$ by

$$F(n) := \sum_{d|n} f(d)$$

Question. Suppose that we know F. How do we recover f ?

Definition 1.26. The Mobius function $\mu : \mathbf{Z}_{\geq 1} \to \mathbf{Z}$ is defined as follows :

Suppose that $n \in \mathbf{Z}_{\geq 1}$ has the prime factorisation

$$n = p_1^{e_1} \cdots p_r^{e_r}$$

then

$$\mu(n) := \begin{cases} 1 & \text{for } n = 1 \\ 0 & \text{if some } e_i > 1 \\ (-1)^r & \text{if } e_1 = e_2 = \cdots = e_r = 1 \end{cases}$$

Lemma 1.33. *For $n \in \mathbf{Z}_{\geq 1}$ we have*

$$\sum_{d|n} \mu(d) = \begin{cases} 1 & \text{if } n = 1 \\ 0 & \text{if } n > 1 \end{cases}$$

Proof. First, we check that the lemma is correct for $n = 1$:

$$\sum_{d|1} \mu(d) = \mu(1) = 1$$

Now suppose that $n \in \mathbf{Z}_{>1}$ has the prime factorisation

$$n = p_1^{e_1} \cdots p_r^{e_r}$$

where $e_1, \dots, e_r \geq 1$. We set $n^* = p_1 \dots p_r$. If $d \mid n$ and $d \nmid n^*$ then d has a prime divisor of multiplicty greater than 1. Then $\mu(d) = 0$. Hence

$$\sum_{d|n} \mu(d) = \sum_{d|n^*} \mu(d)$$

Now we can easily compute

$$\begin{aligned}
\sum_{d|n^*} \mu(d) &= \sum_{d|p_1\cdots p_r} \mu(d) \\
&= \sum_{I\subset\{1,\dots,r\}} \mu\left(\prod_{i\in I} p_i\right) \\
&= \sum_I (-1)^{|I|} = \sum_{k=0}^{r} (-1)^k \binom{r}{k} \\
&= (1-1)^r = 0
\end{aligned}$$

☺

Theorem 1.34 (Mobius inversion formula). *Let $f, F : \mathbf{Z}_{\geq 1} \to \mathbf{C}$ be such that*

$$F(n) = \sum_{d|n} f(d) \tag{1.6}$$

then

$$f(n) = \sum_{d|n} \mu(d) F(n/d) \tag{1.7}$$

Moreover, 1.7 implies 1.6.

Proof. Let d and n be numbers in $\mathbf{Z} \geq 1$ such that d divides n. Then $F(n/d) = \sum_{d'|(n/d)} f(d')$. Therefore

$$\sum_{d|n} \mu(d) F(n/d) = \sum_{d|n} \mu(d) \sum_{d'|(n/d)} f(d')$$

Consider the set S_n of all pairs $(d, d') \in \mathbf{Z}_{\geq 1}$ such that $d \mid n$ and $d' \mid (n/d)$. If n and its divisor d' are fixed, then d runs over all divisors of n/d'.

We can change the order of summation

$$\sum_{d|n} \sum_{d'|(n/d)} \mu(d) f(d') = \sum_{(d,d')\in S_n} \mu(d) f(d') = \sum_{d'|n} \sum_{d|(n/d')} f(d')\mu(d)$$

which gives

$$\sum_{d'|n} \sum_{d|(n/d')} f(d')\mu(d) = f(n)$$

which finished the proof of the inversion formula.

Now we will prove the second statement of the theorem.

Let $F : \mathbf{Z}_{\geq 1} \to \mathbf{C}$ be any function. Set $f(n) := \sum_{d|n} \mu(d) F(n/d)$. Then for $n \in \mathbf{Z}_{\geq 1}$:

$$\sum_{d|n} f(d) = \sum_{d|n} \sum_{d'|d} \mu(d') F(d'/d)$$

We make the change of variables in summation

$$\begin{gathered}
\{(d, d') : d \mid n, d' \mid d\} \simeq \{(d'', d') : d'' \mid n, d' \mid (n/d'')\} \\
(d, d') \mapsto (d/d', d')
\end{gathered}$$

so

$$\begin{aligned}
\sum_{d|n} \sum_{d'|d} \mu(d') F(d'/d) &= \sum_{d''|n} \sum_{d'|(n/d'')} \mu(d') F(d'') \\
&= \sum_{d''|n} F(d'') \sum_{d'|(n/d'')} \mu(d') = F(n)
\end{aligned}$$

☺

1.16 Cyclic sequences

Lecture 7

Definition 1.27. Let A be a set. A linear sequence of length n in alphabet A is an element of A^n :

$$a = (a_1, \dots, a_n), \quad a_k \in A, \; k = 1, \dots, n$$

The number of linear sequences of length n in an alphabet of size r is r^n.

Consider the following equivalence relation on the set of linear sequences

$$(a_1, \dots, a_n) \sim (a_2, a_3, \dots, a_n, a_1)$$

Two linear sequences are equivalent if one of them can be obtained from another by a cyclic shift.

Reminder (Equivalence relation)**.** Let X be a set. A binary relation on X is a subset of $X \times X$. A relation $R \subset X$ is an equivalent relation if it is:

1. Reflexive: $\forall x \in X : (x, x) \in R$.
2. Symmetric: $\forall x, y \in X : (x, y) \in R \Rightarrow (y, x) \in R$.
3. Transitive: $\forall x, y, z \in Z : (x, y) \in R, (y, z) \in R \Rightarrow (x, z) \in R$.

Definition 1.28. A cyclic sequence of length n in alphabet A is an equivalence class of linear sequences with respect to the relation $\sim$.

Example 1.17. There are 8 linear sequences of length 3 in alphabet $\{a, b\}$ and only 4 cyclic sequences.

Proposition 1.35. *The number $T(n, r)$ of cyclic sequences of length n on an alphabet of size r is*

$$T(n, r) = \frac{1}{n} \sum_{d|n} \phi(n/d) r^d$$

where ϕ is the Euler totient function.

Proof.

Definition 1.29. A period of a cyclic sequences $(a_1, \dots, a_n)$ is a minimal number $k \in \{1, \dots, n\}$ such that

$$(a_1, \dots, a_n) = (a_{1+k}, \dots, a_1, \dots, a_k)$$

are equal as linear sequences.

Let $M(d, r)$ be the number of cyclic sequences of length d and period exactly d. The following identity holds

$$r^n = \sum_{d|n} d M(d, r).$$

We have the map

$$\{\text{Linear sequences of length } n \text{ in alphabet } [r]\} \xrightarrow{\pi} \{\text{Cyclic sequences of length } n \text{ in } [r]\}$$

where π is the projection into an equivalency class. The number of preimages of a cyclic sequences under the map π is d, the period of the sequence.

Therefore

$$|L(n, r)| = \sum_{d|n} d|M(d, r)|.$$

Starting with the identity

$$r^n = \sum_{d|n} d M(d, r)$$

we apply the Mobius inversion formula and obtain

$$nM(n,r) = \sum_{d|n} \mu(d/n) r^d. \tag{1.8}$$

Each cyclic sequence has a well defined period d and it corresponds to the unique cyclic sequence of length d and period d. Thus

$$T(n,r) = \sum_{d|n} M(d,r).n \tag{1.9}$$

Now we substitute 1.8 into 1.9:

$$\begin{aligned} T(n,r) &= \sum_{d|n} M(d,r) \\ &= \sum_{d|n} \frac{1}{d} \sum_{d'|d} \mu(d/d') r^{d'} \\ &= \sum_{d'|n} \left(\sum_{d''|(n/d')} \frac{1}{d'd''} \mu(d'') \right) r^{d'} \end{aligned}$$

Now it remains to compute the sum

$$\sum_{d''|(n/d')} \frac{1}{d'd''} \mu(d'')$$

Exercise 1.6. Show that for $n \in \mathbf{Z}_{\geq 1}$, $\sum_{d|n} \frac{1}{d}(\mu(d)) = \frac{\phi(n)}{n}$.

Using this exercise we compute

$$T(n,r) = \sum_{d'|n} \frac{\phi(n/d')}{n} r^{d'}$$

☺

1.17 Partially ordered set (posets)

Definition 1.30. A binary relation on a set A is a subset $R \subseteq A \times A$.

Definition 1.31. A relation is antisymmetric provided $(a,b) \in R$ and $(b,a) \in R$ imply $a = b$.

Definition 1.32. A relation is transitive if $(a,b) \in R$ and $(b,c) \in R$ imply $(a,c) \in R$.

Definition 1.33. A relation is reflexive if $(a,a) \in R$ for all $a \in R$.

Definition 1.34. A partial order on a set A is an antisymmetric, reflexive and transitive relation $R \subseteq A \times A$.

Definition 1.35. A partially ordered set (or poset for short) is a set together with a partial order.

Example 1.18. Let A be a set. The set 2^A is partially ordered by inclusion $\subseteq$.

Example 1.19. The set $\mathbf{Z}_{\geq 1}$ is partially ordered by the relation "d divides n".

Definition 1.36. A partially ordered set $(X, \leq)$ is locally finite if for all $x, y \in X$ the interval

$$[x,y] := \{z \in X : y \leq z \leq x\}$$

is finite.

We say that $0 \in X$ is a zero element if $0 \leq x$ for all $x \in X$.

1.17.1 Mobius inversion formula for posets

Let $(X, \leq)$ be a partially ordered locally finite set with 0. Suppose that $f : X \to \mathbf{C}$ is a function. We define a new function $F : X \to \mathbf{C}$ by

$$F(x) := \sum_{y \leq x} f(y)$$

How do recover f from F?

Theorem 1.36 (Mobius inversion for posets)**.** *Given a partially ordered set X, there is a two-variable function $M : X \times X \to \mathbf{R}$ such that*

$$F(x) = \sum_{y \leq x} f(y) \Leftrightarrow f(x) = \sum_{y \leq x} F(y) M(y, x)$$

M is called the Mobius function of the poset.

Definition 1.37. Given a partially ordered set X, the incidence algebra $A(X)$ is the set of complex-valued functions $f : X^2 \to \mathbf{C}$ satisfying $f(x, y) = 0$ unless $x \leq y$.

$A(X)$ is a vector space over $\mathbf{C}$ with respect to pointwise addition and multiplicavtion by scalars. To make it into an "algebra" we need one more operation:

Definition 1.38. Given $f, g \in A(X)$, their convolution $f * g$ is defined by

$$f * g(x, y) = \sum_{z : x \leq z \leq y} f(x, z) g(z, y)$$

Definition 1.39. The delta function δ is the following element of $A(X)$:

$$\delta(x, y) = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{otherwise.} \end{cases}$$

Lemma 1.37. *A function $f \in A(X)$ has a left and right inverse with respect to the convolution if and ony if $f(x, x) \neq 0$ for all $x \in X$.*

Proof. Given $f \in A(X)$ we find $g \in A(X)$ such that

$$f * g(x, y) = \sum_{x \leq z \leq y} f(x, z) g(z, y) = \delta(x, y)$$

For all $x \in X$ we have that $f * g(x, x) = \delta(x, x) = 1$. Therefore, the condition $f(x, x) \neq 0$ for all $x \in X$ is necessary for the existence of the inverse.

We define $g(x, x) := f(x, x)^{-1}$ for all $x \in X$. To define $g(x, y)$ for $x < y$, we assume by induction, that we have already found all $g(z, y)$ for all z satisfying $x < z \leq y$. Then

$$\delta(x, y) = 0 = \sum_{x \leq z \leq y} f(x, z) g(z, y) - f(x, x) g(x, y) = \sum_{x < z \leq y} f(x, z) g(z, y)$$

We can find $g(x, y)$ from this identity because $f(x, x) \neq 0$ and we know all the terms of the finit sum on the right hand side.

Finally, if $f * g_1 = \delta$ and $g_2 * f = \delta$ then

$$g_2 = g_2 * \delta = g_2 * f * g_1 = (g_2 * f) * g_1 = \delta * g_1 = g_1.$$

Therefore, the left and the right inverses of f coincide. ☺

Definition 1.40. The zeta function $Z(x, y)$ of the poset $(X, \leq)$ is the function

$$Z(x, y) = \begin{cases} 1 & \text{if } x \leq y \\ 0 & \text{otherwise.} \end{cases}$$

The Mobius function $M(x, y)$ is the inverse of the zeta function Z with respect to convolution.

Proof of theorem 1.36. Let $f : X \to \mathbf{C}$ be a function and $F : X \to \mathbf{C}$ is defined by

$$F(x) = \sum_{y \leq x} f(y)$$

Then for a fixed $x \in X$:

$$\begin{aligned}
\sum_{y \leq x} F(y)M(y,x) &= \sum_{y \leq x} M(y,x) \sum_{z \leq y} f(z) \\
&= \sum_{z \leq y \leq x} f(z)M(y,x) \\
&= \sum_{z \leq x} f(z) \sum_{y: z \leq y \leq x} M(y,x) \\
&= \sum_{z \leq x} f(z) \left(\sum_{z \leq y \leq x} Z(z,y)M(y,x) \right) \\
&= \sum_{z \leq x} f(z)(Z * M)(z,x) \\
&= \sum_{z \leq x} f(z)\delta(z,x) = f(x)
\end{aligned}$$

Now we show that the inverse is also true. Let $F : X \to \mathbf{C}$ be a function and we define $f : X \to \mathbf{C}$ by

$$f(x) = \sum_{y \leq x} F(y)M(y,x)$$

foral $x \in X$.

Then

$$\begin{aligned}
\sum_{y \leq x} &= \sum_{y \leq x} \sum_{z \leq y} F(z)M(z,y) \\
&= \sum_{z,y: z \leq y \leq x} F(z)M(z,y) \\
&= \sum_{z \leq y \leq x} F(z)M(z,y)Z(y,x) \\
&= \sum_{z \leq x} F(z) \left(\sum_{z \leq y \leq x} M(z,y)Z(y,x) \right) \\
&= \sum_{z \leq x} F(z)\delta(z,x) = F(x)
\end{aligned}$$

☺

Lemma 1.38. *Let 2^A be the set of subsets of a finite set A. 2^A is partially ordered by inclusion. The Mobius function of 2^A is given by*

$$M(x,y) = (-1)^{|x|-|y|}$$

where $x \subseteq y \subseteq A$.

Proof. We have to show that $M * Z = \delta$. Let $x, y \subseteq A$ such that $x \subseteq y$. We compute

$$\begin{aligned}
M * Z(x,y) = \sum_{z: x \subseteq z \subseteq y} M(x,z)Z(z,y) \\
&= \sum_{x \subseteq z \subseteq y} (-1)^{|x|-|z|} \\
&= \sum_{w \subseteq y \setminus x} (-1)^{|w|} = \begin{cases} 0 & \text{if } |y \setminus x| \geq 1 \\ 1 & \text{if } |y \setminus x| = \varnothing. \end{cases}
\end{aligned}$$

☺

Lemma 1.39. *Consider the set $\mathbf{Z}_{\geq 1}$, partially ordered by the relation "x divides y". then the Mobius function of this poset is:*

$$M(x,y) = \mu(y/x).$$

Proof. We need to show that $M * Z = \delta$.

For $x, y \in \mathbf{Z}_{\geq 1}$ with $x \mid y$ we have

$$\begin{aligned} M * Z(x,y) &= \sum_{\substack{z \in \mathbf{Z}_{\geq 1} \\ x \leq z \leq y}} M(x,z)Z(z,y) \\ &= \sum_{z : x \mid z, z \mid y} \mu(z/x) \\ &= \sum_{d \mid (y/x)} \mu(d) = \begin{cases} 1 & \text{if } \frac{y}{x} = 1 \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

☺

Definition 1.41. A function $f : \mathbf{Z}_{\geq 1} \to \mathbf{C}$ is multiplicative if for all $m, n \in \mathbf{Z}_{\geq 1}$, such that $\gcd(m,n) = 1$, we have Lecture 8

$$f(mn) = f(m)f(n).$$

Exercise 1.7. The Möbius function μ is multiplicative.

Chapter 2

Graph theory

Definition 2.1. A graph G is an ordered pair (V, E) where V is a set of elements called vertices and E is a set of 2-element subsets of V, called edges.

Example 2.1. Let $V = \{1, 2, 3, 4\}$ and $E = \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 1\}\} \subset \binom{V}{2}$. $G = (V, E)$ forms a graph.

Definition 2.2. Let V be a finite set. A complete graph on vertices V (or a clique) is the graph $G = (V, \binom{V}{2})$.

A complete graph with n vertices is denoted K_n.

Definition 2.3. The cycle $C_n = (V, E)$ where $V = \{1, \dots, n\}$ and $E = \{\{1, 2\}, \{2, 3\}, \dots, \{n - 1, n\}, \{n, 1\}\}$ forms a graph.

Let G be a graph.

Definition 2.4. Let $v_1, v_2 \in V$ be vertices of G. If $\{v_1, v_2\} \in E$, we say that v_1 and v_2 are connected by an edge or adjacent.

Definition 2.5. A degree of a vertex $v \in V$ is the number of edges adjacent to it.

Lemma 2.1 (The handshake lemma)**.** *The sum of degrees of all vertices in a (finite) graph G is an even number.*

Proof. The sum of degrees of all vertices is equal to twice the number of edges. ☺

Definition 2.6. A walk is a graph G with a sequence of nodes $v_0, v_1, \dots, v_k$ such that v_0 is adjacent to v_1, v_1 to v_2 and so on; any two consecutive nodes in the sequence must be connected by an edge.

Definition 2.7. A path is a walk such that all its vertices are distinct.

Definition 2.8. A closed walk is a walk such that the first vertex v_0 coincides with the last one v_k.

Definition 2.9. A graph $G = (V, E)$ is connected if for every two vertices $u, v \in V$, there exists a path in G between them.

Definition 2.10. A cycle in a graph $G = (V, E)$ is a sequence of distinct vertices $v_1, \dots, v_r \in V$ with $r \geq 3$ such that v_1 is adjacent to v_2, v_2 to v_3, ..., v_{r-1} to v_r, and v_r to v_1.

Definition 2.11. A tree is a connected graph without cycles.

Definition 2.12. A vertex of degree 1 in a tree is called a leaf.

Lemma 2.2. *Every (finite) tree on $n \geq 2$ vertices has at least two leaves.*

Proof. Consider a path of maximum length, say $v_1, \dots, v_n$. The tree is connected, therefore such path exists and has length of at least 2. The initial point v_1 and the final point v_n must be leaves, otherwise the path can be extended. ☺

Lemma 2.3. *Every tree on n vertices has exactly $n-1$ edges.*

Proof. We prove the lemma by induction on n.

For $n = 1$ we have that the tree has 0 edges. ✓

Suppose that the lemma is true for all trees with n or lesser vertices. Let $T = (V, E)$ be a tree with $n+1$ vertices. By the previous lemma, T has a leave, say $v \in V$. Let e be the unique edge adjacent to v. We define a new graph $T' = (V \setminus \{v\}, E \setminus \{e\})$. This new graph contains no cycles and is connected therefore it is a tree. By the induction hypothesis $|E \setminus \{e\}| = |V \setminus \{v\}| - 1 = n-1$. Therefore, $|E| = |V| - 1$. ☺

2.1 Graph isomorphisms

Definition 2.13. Two graphs $G = (V, E)$ and $G' = (V', E')$ are called isomorphic if there is a bijection $f : V \simeq V'$ such that for all $x, y \in V$:

$$\{x, y\} \in E \Leftrightarrow \{f(x), f(y)\} \in E'$$

Such an f is called an isomorphism of the graphs G and G'. We write $G \simeq G'$.

Definition 2.14. Let $G = (V, E)$ be a graph. A map $f : V \to V$ is an automorphism of G if for all $x, y \in E$:

$$\{x, y\} \in E \Leftrightarrow \{f(x), f(y)\} \in E$$

Question. How many different graphs on vertices $\{1, 2, 3\}$ are there?

Answer. $2^{\binom{3}{2}} = 8$.

Question. How many isomorphism classes on vertices $\{1, 2, 3\}$ are there?

Answer. 4.

Question. What is the number of isomorphism classes of graphs with vertices $\{1, \dots, n\}$?

Answer. The number of different graphs is $2^{\binom{n}{2}}$. Each isomorphism class contains at most $n!$ elements. Therefore,

$$\frac{2^{\binom{n}{2}}}{n!} \leq |\text{isomorphism classes}| \leq 2^{\binom{n}{2}}$$

Notation. An isomorphism class is also called an unlabeled graph.

2.2 Counting labeled trees

Theorem 2.4 (Cayley)**.** *The number of trees on n labeled vertices is n^{n-2}.*

Proof. We will prove this theorem with the help of Prüfer codes. We will define a one-to-one correspondence between the set of all trees on n labeled vertices and the set of sequences of length $n-2$ consisting of numbers in $\{1, \dots, n\}$. As the cardinality of the second set is n^{n-2}, this would imply the statement of the theorem.

Consider Algorithm 1:

- Input: A tree on vertices $\{1, \dots, n\}$.
- Step 1: Find the leaf with the smallest label and write down the number of its neighbor.
- Step 2: Delete this leaf and the only edge adjacent to it.
- Step 3: Repeat until we are left with only two vertices.
- Output: String of labels we have written down.

We can observe that the labels of leaves do not appear in the Prüfer code and that the number of times a label appears in a Prüfer code equals its degree minus one.

We have constructed a map from trees to sequences. Now we have to construct the inverse map. Consider Algorithm 2:

- Input: a sequence $(a_1, \dots, a_{n-2}) \in [n]^{n-2}$.
- Step 1: Draw n nodes.
- Step 2: Make the list $(1, \dots, n)$.
- Step 3: If there are only two numbers left on the list, connect them with an edge and stop. Otherwise, continue to step 4.
- Step 4: Find the smallest number in the list which is not in the sequence. Take the first number in the sequence. Add an edge connecting the nodes whose labels correspond tho those numbers.
- Step 5: Delete the smallest number from the list and the first number in the sequence. This gives a smaller list and shorter sequence. Then return to step 3.

☺

Theorem 2.5. *Algorithm 1 provides a map, name it ψ_1, from the set of trees on vertices $\{1, \dots, n\}$ to the set $\{1, \dots, n\}^{n-2}$.*

$$\psi_1(\text{tree}) = \text{Prufer code of a tree}$$

Algorithm 2 provides a map, name it ψ_2, from the set of sequences $\{1, \dots, n\}^{n-2}$ to the set of trees on vertices $\{1, \dots, n\}$.

$$\psi_2(\text{sequence}) = \text{tree}$$

The maps ψ_1 and ψ_2 are inverse to each other.

$$\psi_1 \circ \psi_2 = \mathrm{Id}_{\text{codes}}, \quad \psi_2 \circ \psi_1 = \mathrm{Id}_{\text{trees}}$$

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Proof. We prove the theorem by induction on n. For $n = 2$, the unique tree has $\varnothing$ as its Prufer code.

Let T be a tree on vertices $\{1, \dots, n\}$. We apply algorithm 1 to T. Let $\bar{a} = (a_1, \dots, a_{n-2})$ be the Prufer code of T. Let T_1 be the tree obtained from T after applying step 2 for the first time. We observe that $(a_2, \dots, a_{n-2})$ is the Prufer code of tree T_1. Note that the vertices of T_1 are labelled by $\{1, \dots, n\} \setminus \{n_0\}$ where n_0 is the smallest leaf in T. Also, the set $\{1, \dots, n\} \setminus \{a_1, \dots, a_{n-2}\}$ is the set of all leaves of T.

Now we apply algorithm 2 to the sequence $\{a_1, \dots, a_{n-2}\}$. The smallest element of the list $\{1, \dots, n\}$ which is not in the sequence $(a_1, \dots, a_{n-2})$ is the smallest leaf n_0 of T. The first iteration of the algorithm adds an edge between nodes n_0 and a_1. By the assumption hypothesis, the algorithm applied to the sequence $(a_2, \dots, a_{n-2})$ will give us the tree T_1 with vertices labelled by the set $\{1, \dots, n\} \setminus \{n_0\}$. This concludes the proof. ☺

2.3 Estimating the number of unlabelled trees

Theorem 2.6. *Let T_n be the number of unlabelled trees with n vertices. Then*

$$2^{n1} \leq \frac{n^{n-2}}{n!} \leq T_n \leq^{n-1}$$

[1]For $n > 30$.

Proof. First, we prove that $T_n \geq \frac{n^{n-2}}{n!}$. We have

$$T_n \geq \frac{|\text{Labelled trees}|}{\text{maximal possible trees in one equivalence class}}$$

Where the denominator equals $n!$.

Now we show that $T_n \leq 4^{n-1}$. Take an unlabelled tree T with n vertices. Choose one vertex of T and call it a root. Embed T into a plane. We start from the root and go counter-clockwise around the tree and write $+$ if the distance to the root increases and $-$ if the distance to the root decreases. At the end of our journey we obtain a sequence in $\{+,-\}^{2n-2}$. We remark that an unlabelled tree can be uniquely reconstructed from this sequence. Since each tree corresponds to at least one, if not more, sequences and each sequence correspond to only on tree, with some to none, then we find

$$T_n \leq 4^{n-1}.$$

☺

Remark 2.7. A difficult result of Otter (1948) says that the number of unlabelled trees with n vertices is

$$\approx (0.5349)n^{-5/2}(2.9557)^n.$$

2.4 Subgraphs and induced subgraphs

Definition 2.15. A subgraph of a graph $G = (V, E)$ is a graph $G' = (V', E')$ such that $V' \subseteq V$ and $E' \subseteq E$.

Definition 2.16. A graph $G' = (V', E')$ is an induced subgraph of $G = (V, E)$ if $V' \subseteq V$ and $E' = E \cap \binom{V'}{2}$.

Definition 2.17. Let $G = (V, E)$ be a graph. An arbitrary tree of the form (V, E'), where $E' \subseteq E$, is called a spanning tree of the graph G.

Lemma 2.8. *Every connected graph contains a spanning tree.*

Proof. Let $G = (V, E)$ be a connected graph. We construct a spanning tree of G by the following algorithm:

- Step 1: Start with an empty subgraph $T = \varnothing$.
- Step 2: Pick an edge $e \in E$ such that $e \notin E(T)$.
- Step 3: Consider the new subgraph T' obtained from T by adding e.
- Step 4: If T' does not contain cycles, set $T := T'$.
- If there is no edge $e \in E$ such that steps 2 and 3 hold, then stop.

Now we check that the output of this algorithm is a spanning tree of G. T contains all vertices of G, is connected, and contains no cycles; therefore T is a spanning tree. ☺

2.4.1 Minimal spanning tree

Definition 2.18. A weighted graph is a graph in which each edge is assigned a numerical weight.

Definition 2.19. We define the weight of a graph as the sum of weights of all its edges.

Problem 2.1. Find a minimum weight spanning tree T for a given weighted connected graph G.

Kruskal's algorithm : a greedy algorithm

- Step 1: Start with an empty graph.
- Step 2: Take all the edges that have not been selected and that would not create a cycle with the already selected edges. Add the one with the smallest weight.
- Step 3: Repeat until the graph is connected and contains all vertices.

Correctness of the algorithm. Let T be the graph obtained as the output of Kruskal's algorithm to a connected weighted graph G. We observe that T contains all vertices of G, is connected and contains no cycles; therefore T is a spanning tree.

Now we show that T has minimal weight. Let F be another spanning tree of G. We want to show that $\mathrm{wt}(F) \geq \mathrm{wt}(T)$. We number the edges of T according to the order we have added them while running the algorithm. Let e be the edge with the smallest number such that $e \in E(T)$ and $e \notin E(F)$. Add e to F, forming a new graph which contains a cycle C.

Since C is not fully contained in T, then C has an edge f that is not an edge of T. If we add the edge e to F, and delete f, we get a third tree H.

We want to show that $\mathrm{wt}(H) \leq \mathrm{wt}(F)$, which is equivalent to $\mathrm{wt}(f) \geq \mathrm{wt}(e)$. Suppose $\mathrm{wt}(f) < \mathrm{wt}(e)$. We have chosen f instead of e since we would form a cycle with the edges of T otherwise. All previously selected edges of T are edges of F, f is an edge of F, then F contains a cycle. However, this is impossible since F is a tree.

This proves that F can not be cheaper than H and H coincides with T after more iterations of the algorithm. ☺

2.5 Graphs and matrices

Let G be a graph. Suppose $|V(G)| = n$ and $|E(G)| = m$.

Definition 2.20. The adjacency matrix of G is the $n \times n$ matrix $A(G)$ given by

$$A_{ij} = \begin{cases} 1 & \text{if } \{v_i, v_j\} \in E \\ 0 & \text{otherwise.} \end{cases}$$

Definition 2.21. The degree matrix of G is the diagonal $n \times n$ matrix $D(G)$ given by

$$D_{ij} = \begin{cases} \deg(v_i) & \text{if } i = j \\ 0 & \text{otherwise.} \end{cases}$$

Definition 2.22. The Laplace matrix of G is $L(G) := D(G) - A(G)$.

Lemma 2.9. *Let G be a graph and $A = A(G)$ be its adjacency matrix. The entries of the matrix $B := A^n$ have the following combinatorial interpretation: B_{ij} is the number of walks of length n starting at v_i and ending at v_j.*

Proof.

$$B_{ij} = \sum_{k_1=1}^{|V(G)|} \sum_{k_2=1}^{|V(G)|} \cdots \sum_{k_{n-1}=1}^{|V(G)|} A_{i,k_1} A_{k_1,k_2} \cdots A_{k_{n-1},j}$$

where

$$A_{i,k_1} A_{k_1,k_2} \cdots A_{k_{n-1},j} = \begin{cases} 1 & \text{if } \{i, k_1, \dots, k_{n-1}, j\} \text{ is a walk} \\ 0 & \text{otherwise.} \end{cases}$$

☺

Let G be a graph. Suppose that $|V(G)| = n$ and $|E(G)| = m$.

Definition 2.23. We say that a graph G has an orientation $\mathcal{O}$ if for every edge $\{v_1, v_2\}$ of G, one of the vertices, say v_1, is the initial vertex and v_2 is the final vertex.

Definition 2.24. The incidence matrix $M(G, \mathcal{O})$ is the $n \times m$ matrix given by

$$M_{ij} = \begin{cases} 1 & \text{if the edge } e_j \text{ has initial vertex } v_i \\ -1 & \text{if the edge } e_j \text{ has final vertex } v_i \\ 0 & \text{otherwise} \end{cases}$$

Lemma 2.10. *We have* $M(G, \mathcal{O})M(G, \mathcal{O})^\mathsf{T} = L(G)$.

Proof. $M(G, \mathcal{O})M(G, \mathcal{O})^\mathsf{T}$ is an $n \times n$ matrix. The ij-th entry of this matrix is

$$\begin{aligned}(M(G, \mathcal{O})M(G, \mathcal{O})^\mathsf{T})_{ij} &= \sum_{k=1}^{m} M_{i,k} M_{j,k} \\ &= \sum_{k: e_k \text{ is adjoint to } v_i \text{ and } v_j} M_{i,k} M_{j,k}\end{aligned}$$

If $i \neq j$, then

$$\sum_{k=1}^{m} M_{i,k} M_{j,k} = \begin{cases} 0 & \text{if } \{v_i, v_j\} \notin E \\ -1 & \text{if } \{v_i, v_j\} \in E \end{cases}$$

If $i = j$ then

$$\sum_{k=1}^{m} M_{i,k} M_{i,k} = \deg(v_i)$$

☺

Theorem 2.11 (Kirchhoff)**.** *Let G be a connected graph on n vertices. Then the rank of the Laplace matrix $L(G)$ is $n-1$.*

Let $0, \lambda_1, \dots, \lambda_{n-1}$ be the eigenvalues of $L(G)$. Then the number of spanning trees of G is

$$\frac{1}{n} \lambda_1 \cdots \lambda_{n-1}.$$

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Example 2.2. Let

$$L(G) = \begin{pmatrix} 2 & -1 & 0 & -1 \\ -1 & 3 & -1 & -1 \\ 0 & -1 & 2 & -1 \\ -1 & -1 & -1 & 3 \end{pmatrix},$$

the characteristic polynomial of this matrix is

$$p_{L(G)}(x) = x^4 - 10x^3 + 32x^3 - 32x$$

with eigenvalues

$$\lambda_1 = 0, \quad \lambda_2 = 2, \quad \lambda_3 = \lambda_4 = 4.$$

Kirchhoff theorem implies that there are 8 spanning trees for G.

Notation. Let

- $M = M(G)$ be an incidence matrix of G,
- $M_0(G)$ be the matrix obtained from $M(G)$ by removing the last row,
- $S \subset E$ be a subset of edges such that $|S| = |V(G)| - 1$.

Then $M_0(S)$ is the submatrix of $M_0(G)$ formed by columns of M_0 indexed by the edges of S.

Lemma 2.12. *Let S be the set of $n-1$ edges of G. If S does not form the set of edges of a spanning tree, then $\det(M_0(S)) = 0$. If S is the set of edges of a spanning tree of G, then $\det(M_0(S)) = \pm 1$.*

Example 2.3. Let

$$M(G, \mathcal{O}) = \begin{pmatrix} -1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & -1 & 0 & -1 \end{pmatrix}$$

and $S = \{e_1, e_3, e_5\}$. Then

$$M_0(S) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

that has determinant $\det(M_0(S)) = -1$.

Proof. First, suppose that S is not the set of edges of a spanning tree. Then, some subset R of S forms the edges of a cycle C in G. Suppose that the cycle C has edges $f_1, \dots, f_s$ in this order. Let $\boldsymbol{w}_1, \dots, \boldsymbol{w}_s$ be the corresponding column vectors of $M_0(S)$. Define

$$k_i := \begin{cases} 1 & \text{if the orientation of } f_i \text{ coincides with the orientation of } C \\ -1 & \text{otherwise.} \end{cases}$$

We have

$$\sum_{i=1}^{s} k_i \boldsymbol{w}_i = \boldsymbol{0}$$

Therefore, $\operatorname{rank}(M_0(S)) < n-1$ meaning that $\det(M_0(S)) = 0$.

Reminder. v_n is the last vertex of G which corresponds to the row removed from M to obtain M_0.

Now suppose that S is the set of edges of a spanning tree T. Let e be an edge of T which is connected to v_n. The column of $M_0(S)$ indexed by e contains exactly one non-zero entry. Remove from $M_0(S)$ the row containing this non-zero entry and the column corresponding to e. We obtain a $(n-2) \times (n-2)$ matrix M_0'. We have that $\det(M_0(S)) = \pm \det(M_0')$.

Let T' be the tree obtained from T by contracting the edge e to a single vertex u. Then M_0' is the matrix obtained from the incidence matrix of T' by removing the row indexed by u. By induction on the number n of vertices we have

$$\det(M_0') = \pm 1$$

which concludes the proof. ☺

Definition 2.25. Let A be a rectangular matrix of size $m \times n$. Suppose $m \leq n$ and S is an m-element subset of $\{1, 2, \dots, n\}$. Then we denote by $A[S]$ the matrix $(A_{ij})_{i=1,\dots,m}$ consisting of columns of A indexed by elements of S.

Theorem 2.13 (Binet-Cauchy)**.** *Let $A, B \in \mathbf{C}^{m \times n}$. If $m \leq n$ then*

$$\det(AB^\mathsf{T}) = \sum_S (\det A[S])(\det B[S])$$

where S runs over all m-element subsets of $\{1, 2, \dots, n\}$.

Proof of 2.11. Let $L_0(G)$ be the matrix obtained from $L(G)$ by removing the last row and the last column.

We have $L_0 = M_0 M_0^\mathsf{T}$. By Binet-Cauchy

$$\begin{aligned}\det(L_0) &= \sum_{\substack{S\subset E\\|S|=n-1}} (\det M_0[S])(\det M_0^\mathsf{T}[S]) \\ &= \sum_{\substack{S\subset E\\|S|=n-1}} (\det M_0[s])^2 \\ &= \sum_S (\pm 1)^2 + \sum_S (0)^2 \\ &= \text{the number of spanning trees in } G.\end{aligned}$$

☺

Lemma 2.14. *Suppose that $A \in K^{n\times n}$ is invertible, $x \in K^n$ and $b \in K$. Then*

$$\det\begin{pmatrix} A & x \\ x & b\end{pmatrix} = (b - x^\mathsf{T} A^{-1} x)\det(A).$$

Proof. Set $\tilde{x} = A^{-1}x$ and $\tilde{b} = b - x^\mathsf{T} A^{-1}x$. Then

$$\det\begin{pmatrix} A & x \\ x & b\end{pmatrix} = \tilde{b}\det(A) = (b - x^\mathsf{T} A^{-1} x)\det(A)$$

☺

Let L be a Laplace matrix of a graph G. We denote by n the number of vertices in $V(G)$.

$$L = \begin{pmatrix} E_{n-1} & 0 \\ -1\cdots-1 & 1\end{pmatrix}\begin{pmatrix} L_0 & (0\cdots 0)^\mathsf{T} \\ 0\cdots 0 & 0\end{pmatrix}\begin{pmatrix} E_{n-1} & (-1\cdots-1)^\mathsf{T} \\ 0\cdots 0 & 1\end{pmatrix}$$

Exercise 2.1. Show that $\det(L_0(G)) = \frac{1}{n}\lambda_1\cdots\lambda_{n-1}$.

Solution. Let $\lambda_1,\ldots,\lambda_n$ be the eigenvalues of L (counted with multiplicity). Since $\operatorname{rank}(L) \le n-1$, we know that at least one of the eigenvalues of L is zero. Suppose that $\lambda_n = 0$.

Let x be a variable. $\det(L + E_n x)$ is a polynomial of degree n in x. We calculate

$$\det(L + E_n x) = \prod_{i=1}^n (x + \lambda_i) = x^n + P_{n-1}x^{n-1} + \cdots + P_1 x + P_0$$

where

$$P_0 = \lambda_1\cdots\lambda_n = 0, \quad P_1 = \sum_{i=1}^n\left(\prod_{j\neq i}\lambda_j\right) = \lambda_1\cdots\lambda_{n-1}.$$

We have

$$L + E_n x = \begin{pmatrix} L_0 + E_{n-1}x & L_0(-1\cdots-1)^\mathsf{T} \\ (L_0(-1\cdots-1)^\mathsf{T})^\mathsf{T} & (-1\cdots-1)L_0(-1\cdots-1)^\mathsf{T} + x\end{pmatrix}$$

the lemma implies that

$$\begin{aligned}\det(L + E_n x) = \det(L_0 + E_{n-1}x)[&(-1\cdots-1)L_0(-1\cdots-1)^\mathsf{T}] \\ &+ x - (-1\cdots-1)L_0^\mathsf{T}(L_0 + E_{n-1}x)^{-1}L_0(-1\cdots-1)^\mathsf{T}]\end{aligned}$$

We have

$$\det(L_0 + E_{n-1}x) = x^{n-1} + \tilde{P}_{n-2}x^{n-2} + \cdots + \tilde{P}_1 x + \tilde{P}_0$$

for some $\tilde{P}_i \in \mathbf{Q}$ and $\tilde{P}_0 = \tilde{\lambda}_1\cdots\tilde{\lambda}_{n-1} = \det(L_0)$. We also consider the Taylor expansion around $x = 0$:

$$\begin{aligned}(-1\cdots-1)L_0(-1\cdots-1)^\mathsf{T}] + x - (-1\cdots-1)L_0^\mathsf{T}(L_0 + E_{n-1}x)^{-1}L_0(-1\cdots-1)^\mathsf{T} &= \\ &= A_0 + A_1 x + A_2 x^2 + o(x^2)\end{aligned}$$

for some $A_0, A_1, A_2 \in \mathbf{R}$. We have $P_0 = \tilde{P}_0 A_0 = 0$, $P_1 = \tilde{P}_0 A_1 + \tilde{P}_1 A_0 = \det L_0 A_1$.

We compute

$$\begin{aligned}(-1 \cdots -1)L_0(L_0 + E_{n-1}x)^{-1}L_0(-1 \cdots -1)^\mathsf{T} &= (-1 \cdots -1)L_0(1 + xL_0^{-1})^{-1}(-1 \cdots -1)^\mathsf{T} \\ &= (-1 \cdots -1)L_0(1 - xL_0^{-1} + x^2 L_0^{-2} + o(x^3))(-1 \cdots -1)^\mathsf{T} \\ &= (-1 \cdots -1)L_0(-1 \cdots -1)^\mathsf{T} - x(-1 \cdots -1)(-1 \cdots -1)^\mathsf{T} + o(x^2)\end{aligned}$$

Therefore,

$$A_1 = (-1 \cdots -1)(-1 \cdots -1)^\mathsf{T} + 1 = n - 1 + 1 = n$$

which means that $\lambda_1 \cdots \lambda_{n-1} = n \det L_0$. ☺

Proof of 2.13. We have

$$A = (A_{ij})_{(i,j)\in[m]\times[n]}, \quad B = (B_{ij})_{(i,j)\in[m]\times[n]}$$

and

$$(AB^\mathsf{T})_{i_1,i_2} = \sum_{j=1}^{n} A_{i_1,j} B_{i_2,j}.$$

We compute

$$\begin{aligned}\det(AB^\mathsf{T}) &= \sum_{\pi\in S_m} \operatorname{sign}(\pi) \prod_{i=1}^{m} (AB^\mathsf{T})_{i,\pi(i)} \\ &= \sum_{\pi\in S_m} \operatorname{sign}(\pi) \prod_{i=1}^{m} \left(\sum_{j=1}^{n} A_{i,j} B_{\pi(i),j} \right) \\ &= \sum_{\pi\in S_m} \operatorname{sign}(\pi) \sum_{(j_1,\dots,j_m)\in[n]^m} \prod_{i=1}^{m} A_{i,j_i} B_{\pi(i),j_i}\end{aligned}$$

and

$$\sum_{\substack{J\subset[n]\\|J|=m}} \det A[J] \det B[J] = \sum_{1\le j_1<\cdots<j_m\le n} \left(\sum_{\pi_1\in S_m} \operatorname{sign}(\pi_1) \prod_{i=1}^{m} A_{i,j_{\pi_1(i)}} \right) \left(\sum_{\pi_2\in S_m} \operatorname{sign}(\pi_2) \prod_{i=1}^{m} B_{i,j_{\pi_2(i)}} \right)$$

Lemma 2.15. *Let $(j_1, \dots, j_m) \in [n]^m$. Suppose that $j_s = j_t$ for some pair of indices $s \neq t$. Then*

$$\sum_{\pi\in S_m} \operatorname{sign}(\pi) \prod_{i=1}^{m} A_{i,j_i} B_{\pi(i),j_i} = 0.$$

Proof. Let $\sigma_{s,t} \in S_m$ be the permutation that interchanges s and t and fixes all other elements of $[m]$. Then for each $\pi \in S_m$ we have

$$\operatorname{sign}(\pi \circ \sigma) = -\operatorname{sign}(\pi).$$

We have

$$\begin{aligned}\prod_{i=1}^{m} (A_{i,j_i} B_{\pi\circ\sigma(i),j_i}) &= \prod_{i\in[m]\setminus\{s,t\}} (A_{i,j_i} B_{\pi\circ\sigma(i),j_i}) \prod_{i\in\{s,t\}} (A_{i,j_i} B_{\pi\circ\sigma(i),j_i}) \\ &= \prod_{i\in[m]\setminus\{s,t\}} (A_{i,j_i} B_{\pi(i),j_i}) A_{s,j_s} B_{\pi(t),j_s} A_{t,j_t} B_{\pi(s),j_t} \\ &= \prod_{i\in[m]\setminus\{s,t\}} (A_{i,j_i} B_{\pi(i),j_i}) A_{s,j_s} B_{\pi(t),j_t} A_{t,j_t} B_{\pi(s),j_s} \\ &= \prod_{i=1}^{m} (A_{i,j_i} B_{\pi(i),j_i}).\end{aligned}$$

thus

$$\operatorname{sign}(\pi)\prod_{i=1}^{m} A_{i,j_i} B_{\pi(i),j_i} = -\operatorname{sign}(\pi\circ\sigma)\prod_{i=1}^{m} A_{i,j_i} B_{\pi\circ\sigma(i),j_i}$$

Since the map $\pi \to \pi\circ\sigma$ is a bijection on S_m then

$$\sum_{\pi\in S_m}\operatorname{sign}(\pi)\prod_{i=1}^{m} A_{i,j_i} B_{\pi(i),j_i} = 0.$$

☺

We use this lemma

$$\begin{aligned}
\det(AB^{\mathsf{T}}) &= \sum_{\pi\in S_m}\operatorname{sign}(\pi)\prod_{i=1}^{m}(AB^{\mathsf{T}})_{i,\pi(i)} \\
&= \sum_{\pi\in S_m}\operatorname{sign}(\pi)\prod_{i=1}^{m}\left(\sum_{j=1}^{n} A_{i,j}B_{\pi(i),j}\right) \\
&= \sum_{\pi\in S_m}\operatorname{sign}(\pi)\sum_{(j_1,\dots,j_m)\in[n]^m}\prod_{i=1}^{m} A_{i,j_i}B_{\pi(i),j_i} \\
&\overset{\text{lem}}{=} \sum_{\pi\in S_m}\operatorname{sign}(\pi)\sum_{(j_1,\dots,j_n)\in[n]^m}\prod_{i=1}^{m} A_{i,j_i}B_{\pi(i),j_i} \\
&= \sum_{\pi\in S_m}\operatorname{sign}(\pi)\sum_{1\le j_1<\dots<j_m\le n}\sum_{\pi'\in S_m}\prod_{i=1}^{m} A_{i,j_{\pi'(i)}}B_{\pi(i),j_{\pi'(i)}} \\
&= \sum_{1\le j_1<\dots<j_m\le n}\sum_{\pi\in S_m}\sum_{\pi'\in S_m}\operatorname{sign}(\pi)\left(\prod_{i=1}^{m} A_{i,j_{\pi'(i)}}\right)\left(\prod_{i=1}^{m} B_{i,j_{\prime_{\pi'\circ\pi^{-1}(i)}}}\right)
\end{aligned}$$

we make a change of variables $\pi' = p_1$ and $\pi'\circ\pi^{-1} = \pi_2$. Then $\pi = \pi_2^{-1}\circ\pi_1$ and $\operatorname{sign}(\pi) = \operatorname{sign}(\pi_1)\operatorname{sign}(\pi_2)$.

$$\begin{aligned}
&= \sum_{1\le j_1<\dots<j_m\le n}\sum_{\pi_1\in S_m}\sum_{\pi_2\in S_m}\operatorname{sign}(\pi_1)\operatorname{sign}(\pi_2)\left(\prod_{i=1}^{m} A_{i,j_{\pi_1(i)}}\right)\left(\prod_{i=1}^{m} B_{i,j_{\pi_2(i)}}\right) \\
&= \sum_{1\le j_1<\dots<j_m\le n}\left(\sum_{\pi_1\in S_m}\operatorname{sign}(\pi_1)\left(\prod_{i=1}^{m} A_{i,j_{\pi_1(i)}}\right)\right)\left(\sum_{\pi_2\in S_m}\operatorname{sign}(\pi_2)\left(\prod_{i=1}^{m} B_{i,j_{\pi_2(i)}}\right)\right) \\
&= \sum_{1\le j_1<\dots<j_m\le n}\det(A[\{j_1,\dots,j_m\}])\det(B[\{j_1,\dots,j_m\}]) \\
&= \sum_{\substack{J\subset[n]\\|J|=m}}\det(A[J])\det(B[J])
\end{aligned}$$

which concludes the proof. ☺

2.6 Combinatorial probability

Lecture 11

Definition 2.26. A finite probability space is a pair (Ω, P) where Ω is a finite set and $P : 2^{\Omega} \to [0,1]$ is a function assigning a number from the interval $[0,1]$ to every subset of Ω such that :

1. $P(\varnothing) = 0$,
2. $P(\Omega) = 1$,
3. $P(A\cup B) = P(A) + P(B)$ for any two disjoint set $A, B \subseteq \Omega$.

Probability theory to set theory dictionary The set Ω can be thought as the set of all possible outcomes of some random experiment.

- Elements of Ω are called elementary events.
- Subsets of Ω are called events.

Let $w \in \Omega$, $A, B \subseteq \Omega$:

- $w \in A$ means that A occurred.
- $w \in A \cap B$ means that both events A and B occurred.
- $A \cap B = \varnothing$ if events A and B are incompatible.
- $P(A)$ denotes the probability of event A.

Example 2.4 (A random sequence of 0s and 1s)**.** Let $\Omega = \{0,1\}^n$ = all n-term sequences of 0s and 1s, $|\Omega| = 2^n$. For $A \subseteq \Omega$, $P(A) := |A|/|\Omega|$.

Consider a set $A \subseteq \Omega$ defined by $A := \{(w_1, \dots, w_n) : w_1 = 1\}$. A is the event: "the first element of a random sequence is 1".

Example 2.5 (A random permutation)**.** $\Omega = S_n$ is the set of all permutations of the set $\{1, \dots, n\}$. For $A \subset \Omega$, $P(A) = \frac{|A|}{n!}$.

Reminder. Hatcheck lady problem.

Example 2.6 (A random graph)**.** Let $\Omega = \mathcal{G}_n$ be the set of all possible labelled graphs on vertex set $V = \{1, \dots, n\}$. For $A \subset \mathcal{G}_n$, $P(A) = |A|2^{-\binom{n}{2}}$.

Proposition 2.16. *A random graph is almost neve a tree, i.e.*

$$\lim_{n\to\infty} P(\text{"a graph in } \mathcal{G} \text{ is a tree"}) = 0$$

Proof. We have $P(T_n) = \frac{|T_n|}{|\mathcal{G}_n|}$. By Cayley's theorem, $|T_n| = n^{n-2}$, therefore

$$\lim_{n\to\infty} \frac{n^{n-2}}{2^{n(n-1)/2}} = \lim_{n\to\infty} e^{-\ln(2)n(n-1)/2+(n-2)\ln(n)} = 0.$$

☺

Definition 2.27. Two events A, B in probability space (Ω, P) are called independent if

$$P(A \cap B) = P(A)P(B).$$

Example 2.7. Let $\Omega = \{0,1\}^n$ be the probability space of random sequences. Consider 2 events :

- Event A is defined as "the first element of a sequence is 1".
- Event B is defined as "the second element of a sequence is 1".

Events A and B are independent.

Example 2.8 (Non-example)**.** Let $\Omega = \{0,1\}^n$ be the probability space of random sequences.

- Event A is defined as "all elements of the sequence are 1".
- Event B is defined as "all elements of the sequence are 0".

We have that $P(A) > 0$ and $P(B) > 0$, however $P(A \cap B) = P(\varnothing) = 0$. Therefore, these events are not independent.

Definition 2.28. Events $A_1, \dots, A_n \subseteq \Omega$ are independent if and only if for each subset of indices $I \subseteq \{1, \dots, n\}$ we have

$$P\left(\bigcap_{i\in I} A_i\right) = \prod_{i\in I} P(A_i).$$

2.6.1 Random variables and their expectations

Definition 2.29. Let (Ω, P) be a finite probability space. A random variable on Ω is any map $f : \Omega \to \mathbf{R}$.

Example 2.9. Let $(\{0,1\}^n, |A|/|\Omega|)$ be the probability space of n-term random sequences of 1s and 0s. $f_1(w)$ that returns the number of 1s in a random sequence w is a variable

Example 2.10. The number of fixed points in a random sequence.

Example 2.11. The number of triangles in a random graph.

Definition 2.30. Let (Ω, P) be a finite probability space, and let f be a random variable on it. The expectation of f is a real number $\mathbb{E}(f)$ defined by the formula

$$\mathbb{E}(f) = \sum_{w \in \Omega} P(\{w\}) f(w).$$

Example 2.12. Let f_1 be the number of 1s in a random n-term sequence of 1s and 0s. Then

$$\begin{aligned}\mathbb{E}(f_1) &= \sum_{w \in \{0,1\}^n} f_1(w) 2^{-n} \\ &= 2^{-n} \sum_{k=0}^{n} k \binom{n}{k} \\ &= 2^{-n} \frac{d}{dx}(1+x)^n|_{x=1} \\ &= 2^{-n} n 2^{n-1} = \frac{n}{2}.\end{aligned}$$

Question. Can we find an easier way to compute $\mathbb{E}(f_1)$?

Definition 2.31. Let $A \subseteq \Omega$ be an event in a probability space (Ω, P). The indicator of event A is the random variable $I_A : \Omega \to \{0,1\}$ defined as

$$I_A(w) = \begin{cases} 1 & \text{for } w \in A, \\ 0 & \text{for } w \notin A. \end{cases}$$

Lemma 2.17. *For any event A, we have $\mathbb{E}(I_A) = P(A)$.*

Proof.

$$\mathbb{E}(I_A) = \sum_{w \in \Omega} I_A(w) P(\{w\}) = \sum_{w \in A} P(\{w\}) = P(A).$$

☺

Theorem 2.18 (Linearity of expectation)**.** *Let f, g be arbitrary random variables on a finite probability space (Ω, P) and let $\alpha \in \mathbf{R}$. Then:*

$$\mathbb{E}(\alpha f) = \alpha \mathbb{E}(f) \quad \textit{and} \quad \mathbb{E}(f+g) = \mathbb{E}(f) + \mathbb{E}(g).$$

Example 2.13 (Reprise of 2.12: an alternative method)**.** For $i = 1, \dots, n$, let A_i be the event "the i-th element of the random sequence is 1". Then $P(A_i) = \frac{1}{2}$. For each $w \in \{0,1\}^n$, we have

$$f_1(w) = I_{A_1}(w) + I_{A_2}(w) + \cdots + I_{A_n}(w).$$

Therefore

$$\begin{aligned}\mathbb{E}(f_1) &= \mathbb{E}(I_{A_1}) + \cdots + \mathbb{E}(I_{A_n}) \\ &= P(A_1) + \cdots + P(A_n) \\ &= \frac{n}{2}.\end{aligned}$$

2.6.2 Probabilistic method

Theorem 2.19. *Let (Ω, P) be a finite probability space and let $f : \Omega \to \mathbf{R}$ be a random variable. If $\mathbb{E}(f) = m$, then there exists at least one elementary event w_1 such that $f(w_1) \geq m$.*

Analogously, there exists at least one elementary event w_2 such that $f(w_2) \leq m$.

Application : Existence of large bipartite subgraphs

Theorem 2.20. *Let G be a graph with an even number $2n$ of vertices and with $m > 0$ edges. Then the set $V = V(G)$ can be divided into two disjoint n-element subsets A and B in such a way that more than $m/2$ edges go between A and B.*

Proof. Consider the probability space (Ω, P) where $\Omega = \binom{V}{n}$ with the probability measure $P(S) = |S|/|\Omega|$ for $S \subseteq \Omega$.

Let $A \in \Omega$ be a random n-element subset of $V(G)$. Define $B := V(G)P\{A\}$. Consider the following random variable

$$X(A) := |\text{edges between } A \text{ and } B| = |\{\{a,b\} : a \in A, b \in B, \{a,b\} \in E(G)\}|.$$

Let us compute $\mathbb{E}(X)$. For $e = \{u, v\} \in E(G)$ we define the event $C_e := \{A \in \Omega : |A \cap e| = 1\}$. We have

$$X = \sum_{e \in E(G)} I_{C_e}$$

and therefore

$$\mathbb{E}(X) = \sum_{e \in E(G)} \mathbb{E}(I_{C_e}) = \sum_{e \in E(G)} P(C_e).$$

We compute

$$P(C_e) = \frac{2\binom{2n-2}{n-1}}{\binom{2n}{n}} = \frac{n}{2n-1} > \frac{1}{2}.$$

Thus $\mathbb{E}(X) > \frac{m}{2}$. By theorem 2.19 then $X(A) > \frac{m}{2}$ for some $A \in \Omega$. ☺

Application: Turan's theorem

Definition 2.32. Let G be a graph. A set $S \subseteq V(G)$ is an independent set if no two vertices of S are connected by an edge. $\alpha(G)$ represents the size of the largest independent set of vertices in the graph G.

Theorem 2.21 (Turan)**.** *For every graph G we have*

$$\alpha(G) \geq \frac{|V(G)|^2}{2|E(G)| + V(G)}.$$

Lemma 2.22. *For any graph G we have*

$$\alpha(G) \geq \sum_{v \in V(G)} \frac{1}{\deg(v) + 1}.$$

Proof. Suppose that the vertices of G are numbered $1, \ldots, n$. Pick a random permutation π of the vertices. Define the set $M(\pi) \subset V(G)$ by

$$M(\pi) := \{v \in V : \text{all neighbours } u \text{ of } v \text{ satisfy } \pi(u) > \pi(v)\}.$$

$M(\pi)$ is an independent set of G. Therefore, $|M(\pi)| \leq \alpha(G)$ for any permutation π and $\mathbb{E}(|M(\pi)|) \leq \alpha(G)$.

Now, we calculate the expected size of M. Let $v \in V$, A_v is the event "$v \in M(\pi)$". Then

$$P(A_v) = \frac{1}{\deg(v) + 1}$$

and
$$|M(\pi)| = \sum_{v\in V} I_{A_v}(\pi).$$
Now
$$\mathbb{E}(|M|) = \sum_{v\in V} \mathbb{E}(I_{A_v}) = \sum_{v\in V} P(A_v) = \sum_{v\in V} \frac{1}{\deg(v)+1}.$$
Since $\alpha(G) \geq \mathbb{E}(|M|)$ this finishes the proof. ☺

Proof of 2.21. Let $|V(G)| = n$ and $d_1, \dots, d_n$ be the degrees of vertices of G. Then
$$\sum_{i=1}^{n} d_i = 2|E(G)|$$
and
$$\sum_{i=1}^{n} \frac{1}{d_i+1} \geq \frac{n^2}{d_1+\dots+d_n+n} = \frac{n^2}{2|E|+n}$$
due to the inequality between the arithmetic mean and harmonic mean. ☺

Lecture 12

2.7 Ramsey numbers

Definition 2.33. The Ramsey number $R(k,\ell)$ is

$R(k,\ell) = \min\{n :$ any graph on n vertices contains a clique of size k or an independent set of size $\ell\}$

Theorem 2.23 (Ramsey)**.** *The Ramsey number $R(k,\ell)$ is always finite.*

Proof. We will prove this statement by induction on $k+\ell$. The base of induction is the case $R(2,m) = R(m,2) = m$. For the step of induction we use the following lemma.

Lemma 2.24.
$$R(k,\ell) \leq R(k-1,\ell) + R(k,\ell-1).$$

Proof. Suppose that $R(k,\ell-1)$ and $R(k-1,\ell)$ are finite. Let G be a graph on $R(k,\ell-1)+R(k-1,\ell)$ vertices. Let v be a vertex of G. The degree of v is either greater or equal than $R(k-1,\ell)$ or less than $R(k-1,\ell)$.

If $\deg(v) \geq R(k-1,\ell)$ we consider the subgraph G' of G induced on the neighborhood of v. G' has at least $R(k-1,\ell)$ vertices. Therefore G' contains either a clique of size $k-1$ or an independent set of size ℓ. Therefore G contains either a clique of size k or an independent set of size ℓ.

If $\deg(v) < R(k-1,\ell)$, then v is not connected to at least $R(k,\ell-1)$ points. We consider the subgraph G'' of G induced on the vertices not connected to v. G'' contains either a clique of size k or an independent set of size $\ell-1$. Therefore, G contains either a clique of size k or an independent set of size ℓ. ☺

Definition 2.34. Let G be a graph and v a vertex of G. The neighborhood of v in G is the set of all vertices of G adjacent to v.

The conclusion of the proof is left as an exercise. ☺

Corollary 2.25.
$$R(k,\ell) \leq \binom{k+\ell-2}{k-1}.$$

Theorem 2.26. *For any $k \geq 3$,*
$$R(k,k) > 2^{k/2-1}.$$

Proof. It suffices to prove that there exists a graph $G(V, E)$ such that $V \geq 2^{k/2-1}$ and G contains no cliques and no independent sets of size k.

Consider the finite probability space $\mathcal{G}\left(n, \frac{1}{2}\right)$ of graphs on n vertices where the probability of each edge is $\frac{1}{2}$. For any fixed k vertices, the probability that they form either a clique or an independent set is of $2^{-\binom{k}{2}}$. Let Q be the event "A random graph in $\mathcal{G}\left(n, \frac{1}{2}\right)$ contains a clique or an independent set", we have

$$\begin{aligned} P(Q) &= P\left(\bigcup_{S \text{ is a subset of size } k \text{ in } [n]} \{S \text{ forms a clique}\} \cup \{S \text{ forms an independent set}\}\right) \\ &\leq 2\binom{n}{k} 2^{-\binom{k}{2}}. \end{aligned}$$

Suppose that $n \leq 2^{k/2-1}$. Then

$$\begin{aligned} 2\binom{n}{k} 2^{-k(k-1)/2} &\leq 2n^k 2^{-k(k-1)/2} \\ &\leq 2 \cdot 2^{(k/2-1)k} \cdot 2^{-k(k-1)/2} \\ &= 2^{1+k^2/2-k-k^2/2+k/2} \\ &= 2^{1-k/2} < 1. \end{aligned}$$

Therefore, there exists a graph with n vertices and containing neither a clique nor an independent set of size k. ☺

2.8 Erdos-Ko-Rado or Sunflower Theorem

Definition 2.35. A family $\mathcal{F}$ of sets is intersecting if for all $A, B \in \mathcal{F}$, we have that $A \cap B \neq \varnothing$.

Theorem 2.27 (Erdos-Ko-Rado)**.** *If $|X| = n \geq 2k$ and $\mathcal{F}$ is an intersecting family of k-element subsets of X, then*

$$|\mathcal{F}| \leq \binom{n-1}{k-1}.$$

Example 2.14 (Construction of $\mathcal{F}$)**.** Suppose $X = \{1, \dots, n\}$, we construct

$$\mathcal{F}_k = \{A \subseteq X : 1 \in A \wedge |A| = k\}.$$

Then $\mathcal{F}_k$ is an intersecting family where each set has size k and $|\mathcal{F}_k| = \binom{n-1}{k-1}$.

Lemma 2.28. *Consider $X = \{0, 1, \dots, n-1\}$ with addition modulo n and define*

$$A_s = \{s, s+1, \dots, s+k-1\} \subseteq X$$

for $0 \leq s < n$. Then for $n \geq 2k$, any intersecting family $\mathcal{F} \subseteq \binom{X}{k}$ contains at most k of the sets A_s.

Proof. Without loss of generality, we may assume that $A_0 \in \mathcal{F}$. A set A_s intersects with A_0 only for $s = 0$ or $s \in \{1, \dots, k-1, -1, \dots, -k+1\} \mod n$. We can divide these numbers into two pairs $(j, j-k)$ for $j = 1, \dots, k-1$. Note that $A_j \cap A_{j-k} = \varnothing$. Therefore, only one of these two sets can be contained in $\mathcal{F}$. ☺

Proof of theorem 2.27. We assume that $X = \{0, 1, \dots, n-1\}$ and $\mathcal{F} \subseteq \binom{X}{k}$ is an intersecting family. For a permutation $\sigma : X \to X$, we define

$$\sigma(A_s) = \{\sigma(s), \sigma(s+1), \dots, \sigma(s+k-1)\}$$

with addition modulo n.

The lemma implies that if we choose randoms s and σ independently and uniformly, then

$$P[\sigma(A_s) \in \mathcal{F}] \leq \frac{k}{n}.$$

This choice of $\sigma(A_s)$ is equivalent to a random choice of k-element subset of X. Therefore

$$P(\sigma(A_s) \in \mathcal{F}) = \frac{|\mathcal{F}|}{\binom{n}{k}}.$$

Now we estimate

$$|\mathcal{F}| = \binom{n}{k} P(\sigma(A_s) \in \mathcal{F}) \leq \binom{n}{k}\frac{k}{n} = \binom{n-1}{k-1}.$$

☺

2.9 Bipartite graphs and matchings

Definition 2.36. A bipartite graph is a graph G such that the set of its vertices can be partitioned into two disjoint parts $V(G) = A \sqcup B$ such that every edge of G connects a vertex in A with a vertex in B.

Lemma 2.29. *A graph G is bipartite if and only if it does not contain a cycle of odd length.*

Proof. $\Rightarrow$ direction: Suppose that G contains a cycle C of odd length. Then two subsequent vertices of C belong to the same part. A contradiction.

$\Leftarrow$ direction: Suppose that G contains no odd cycles. We will show that G is bipartite. Let $v, w \in V(G)$ be two vertices.

Claim 2.30. *If G has no odd cycles, then all paths between v and w have the same length modulo 2.*

We fix $v_0 \in V(G)$. We define $A, B \subset V(G)$ as follows:

- A is the set of vertices with even distance from v_0.
- B is the set of vertices with odd distance from v_0.

No edges of G go between A and A or between B and B. Is it true that $V(G) = A \sqcup B$? No. So we repeat the same procedure for all connected components of G. ☺

Definition 2.37. Let $G = (V, E)$ be a graph. A subset $E_0 \subseteq E$ of edges such that $e \cap f = \varnothing$ for all e, $f \in E_0$ is called a matching in G.

Definition 2.38. A perfect matching is a matching that covers all the vertices of a graph. That is, each vertex is adjoint to exactly one edge of the matching.

Matchings in bipartite graphs

Let G be a bipartite graph such that $V(G) = A \sqcup B$ is a partition. Does G have a perfect matching? We stablish the necessary conditions:

- $|A| = |B|$.
- Let $X \subseteq A$ be a subset. We define

  $$Y(X) := \{y \in B : y \text{ is connected to some vertex in } X\}.$$

  A necessary condition for the existence of a perfect matching is that $|Y(X)| \geq |X|$.

Theorem 2.31 (Konig-Hall)**.** *Let G be a bipartite graph with two parts A and B. Suppose that the following two conditions hold:*

- $|A| = |B|$.
- *For each set $X \subseteq A$ the subset*

$$Y(X) := \{y \in B : y \text{ is connected to some vertex in } X\},$$

  satisfies $|Y(X)| \geq |X|$.

Then G has a perfect matching.

Proof. We say that a bipartite graph is "good" if it satisfies both conditions.

Suppose that a bipartite graph G is good. Is it true that every subset $Y \subseteq B$ is connected to at least $|Y|$ vertices in A? We claim that yes.

Indeed, suppose that G is good and there exists a subset $Y \subseteq B$ which is connected to less than $|Y|$ vertices in A. We denote by $X(Y)$ the set of all vertices connected to Y. Note that the set $A \setminus X(Y)$ is connected only to the vertices in the set $B \setminus Y$. By out assumption, $|Y| > |X(Y)|$ and therefore

$$|B \setminus Y| < |A \setminus X(Y)|.$$

This contradicts our assumption that G is good.

We have to prove that every good bipartite graph G has a perfect matching. We will prove this y induction on the number of vertices. For a bipartite graph with two vertices, the statement is obvious. We will prove the following claim:

Claim 2.32. *If a good graph has more than two vertices it can be divided into two good graphs.*

First attempt Let $a \in A$ and $b \in B$ be two vertices connected by an edge. Set $A_1 := \{a\}$ and $B_1 := \{b\}$ and $A_2 := A \setminus \{a\}$ and $B_2 := B \setminus \{b\}$. Then G_1 is defined as the subgraph of G induced by $A_1 \cup B_1$ and G_2 the subgraph of G induced by $A_2 \cup B_2$.

The subgraph G_1 is obviously good. Is G_2 also good?

Suppose that the graph G_2 is not good and the set A_2 contains a subset X such that the set Y of all vertices in B_2 connected to X contains less than $|X|$ elements. Since G is good, we know that the number of vertices in B connected to X is exactly $|X|$. This is possible if and only if $Y(X) = Y \cup \{b\}$.

Second attempt We define $\tilde{A}_1 := X$ and $\tilde{B}_1 := Y \cup \{b\}$. X is the smallest subset such that the set Y of all vertices in B_2 connected to X contains less than $|X|$ elements. Therefore, the bipartite graph induced of $\tilde{A}_1 \sqcup \tilde{B}_1$ is good. Is the graph induced by $\tilde{A}_2 = A \setminus X$ and $\tilde{B}_2 = B \setminus Y \setminus \{b\}$ good?

The vertices of $\tilde{B}_2$ are connected only to the vertices of $\tilde{A}_2$. Since the graph G is good, every subset $\tilde{Y} \subseteq \tilde{B}_2$ is connected to at least $|\tilde{Y}|$ vertices. Also

$$|\tilde{B}_2| = |B| - |Y(X)| = |A| - |X| = |\tilde{A}_2|.$$

Therefore, the graph $\tilde{G}_2$ is also good. Note that $X \subsetneq A$. This finishes the proof of claim 2.32 and also the proof of the theorem by induction on the number of vertices. ☺